

**ISHA: Integrated Safety and Hazard Alert System for Home Emergencies**

**A Capstone Project  
Presented to the Faculty of the  
Information and Communications Technology Program  
STI College Caloocan**

**In Partial Fulfilment  
of the Requirements for the Degree  
Bachelor of Science in Computer Engineering**

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## **ENDORSEMENT FORM FOR ORAL DEFENSE**

**TITLE OF RESEARCH:**            **ISHA: Integrated Safety and Hazard Alert  
System for Home Emergencies**

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## INTRODUCTION

In an era of rapid technological growth, the incorporation of smart technology into everyday operations is transforming industries throughout the world. The Internet of Things (IoT) represents this transformation by allowing objects to interact effortlessly and share data in real time. This connectedness enables firms to increase productivity, make more informed decisions, and respond quickly to emerging difficulties. For instance, in the field of emergency management, reaction is crucial to ensuring public safety and security. Effective emergency management and response efforts can help save lives since emergencies may occur unexpectedly and typically require immediate action. In many circumstances, a few residents lack knowledge to handle emergency situations causing delay in responding can make the situation extremely serious, if not fatal. IoT-enabled fast communication allows for quick and decisive action in response to a crisis, preventing it from worsening. Effective emergency management and response efforts can help save lives since emergencies may occur unexpectedly and typically require an immediate response. Having excellent plans and responses might possibly reduce the impact of dangers while encouraging a quicker recovery.

The integration of the Internet of Things (IoT) in disaster management and response. Enabling adequate and improved coordination reduces the time required to respond to any emergency response. The usage of interconnected equipment and technology allows an emergency to be handled efficiently. This includes the use of sensors and real-time data monitoring, as well as communication and cooperation with emergency personnel.

The aim of this study is to look into the use of the internet of things (IoT) in emergency communication. The widespread adoption of Internet of Things-based technologies in emergency communication systems is an important step toward public safety. The purpose of this article is to provide an overview of the internet of things' (IoT) applications and benefits in public safety and emergency response. Utilizing real-time data and analytics, the systems can enhance situational awareness and encourage collaboration among residents and local responders so that our communities can have faster, more efficient responses to emergencies, leading to better outcomes in crisis situations.

## **Background of the problem**

Emergencies always required immediate and effective communication to prevent potential harm, but many communities struggled or could not provide faster emergency communication. The development of IoT technology opened a new opportunity for enhancing emergency communication through systems that could communicate with emergency responders. Through using IoT, emergency responses were delivered faster so that they could save lives and minimize potential damages.

This paper aimed to address this specific problem by providing an integrated platform that linked emergency detection devices to residents. The system focused on the absolute collection and transmission of data using communication modules and sensors, as well as software for real-time monitoring.

The importance of this research lay in its potential to improve community safety and preparedness by providing early warnings that could reduce response time. While current systems were limited by connectivity and electricity issues, this review explored ways to address these challenges, contributing to safer, more efficient communities.

## **General Problem**

Many homes lacked a quick and reliable way to detect dangers like fires, gas leaks, smoke, and earthquakes. As a result, people might not have known there was a problem until it was too late, which could cause delays in responding to an emergency. These delays could have serious consequences, such as injuries, loss of life, and major property damage. Existing systems often failed to detect hazards early enough or did not warn people in time. Without a dependable device that could alert families to these risks immediately, they were left vulnerable. This paper aimed to solve that by developing a system that combined different types of sensors, notification tools, and monitoring technology. The idea was to give people fast, clear alerts so they could react quickly, stay safe, prevent damage, and reduce the risk of harm during an emergency.

### **Specific Problems:**

- 1. How to develop a device capable of detecting earthquakes, fire hazards, gas leaks, and smoke?**

Developing a multi-hazard detection device required integrating advanced sensors that could detect several hazards simultaneously. This device was designed to provide real-time detection of earthquakes, fire hazards, gas leaks, and smoke.

- 2. How to develop a device capable of notifying residents when earthquakes, fire hazards, gas leaks, and smoke are detected?**

To develop a device capable of notifying residents about earthquakes, fire hazards, gas leaks, and smoke detection, this involved integrating various sensors and technologies to ensure reliability and timely alerts.

- 3. How to develop a device that is capable of preventing fire in its early stage by using YOLOv8n and automated water?**

The development of YOLOv8n real-time object detection required the integration of advanced sensors and machine learning algorithms to accurately detect fires in a residential area. The system was optimized to use the mist system to extinguish fires, ensure minimal damage, and enhance safety for residents.

## **Overview of the current state of the technology**

Emergencies like earthquakes, fires, gas leaks, and smoke-related incidents posed serious risks to lives and property. Over the years, various technologies had been developed to address these problems, but gaps still existed in delivering timely and efficient responses. Early solutions, such as basic smoke alarms and manual fire warning systems, provided some protection but often fell short in speed and reliability. Earthquake monitoring tools began with analog instruments that offered limited warning times, while early gas leak detection relied heavily on manual checks, leading to delays in identifying hazards.

Today's technologies introduced significant improvements to address these challenges. Environmental sensors were now widely used to monitor changes in air quality, temperature, and seismic activity in real time. Smoke detectors had advanced to use dual-sensor systems, combining photoelectric and ionization methods to detect both slow-burning and fast-flaming fires. Gas leak detectors had also become more accurate with the use of semiconductor and infrared technology, enabling early alerts to harmful gases like carbon monoxide or methane. Additionally, modern earthquake detection systems used advanced seismic sensors to detect early tremors and sent warnings seconds before stronger waves hit, providing critical time for safety measures.

Despite these advancements, there were still limitations in how these systems functioned. Many existing devices only addressed specific hazards and were not designed to work as an all-in-one solution for multiple types of emergencies. Notification systems were not always reliable or accessible, and delays in communication could still occur. Furthermore, current devices were often limited in user-friendliness, making it difficult for certain groups, such as the elderly or people with disabilities, to respond effectively during emergencies. These issues highlighted the need for an integrated and accessible device that could monitor multiple hazards and deliver timely and reliable notifications to residents. This research aimed to fill these gaps by creating a device that could detect and alert residents to earthquakes, fires, gas leaks, and smoke in a unified system. The focus was on ensuring the device delivered clear and accessible notifications while providing.

comprehensive monitoring to help residents act quickly and safeguard their homes. By addressing the shortcomings of current solutions, this research will contribute to improving safety and emergency preparedness in homes.

### **Objectives of the study**

The general objective of this study was to develop an integrated emergency detection and alert system that could monitor multiple hazards such as earthquakes, fire hazards, gas leaks, and smoke, while providing clear and timely notifications to residents. This system aimed to enable quick and effective responses, ensuring the safety and security of homeowners in emergency situations. To achieve this, the study focused on several specific objectives: First, it aimed to design a device capable of detecting and monitoring various emergency hazards, including earthquakes, fire hazards, gas leaks, and smoke, using advanced environmental sensors. Second, it sought to develop a reliable communication system that could send clear, timely, and accessible notifications to residents, ensuring that they could respond quickly during an emergency. Third, the study intended to create a user-friendly interface that made it easy for all residents, including vulnerable groups such as the elderly or those with disabilities, to interpret and act on the alerts. Fourth, the research aimed to integrate multiple hazard detection capabilities into a single unified system, offering a comprehensive solution rather than addressing individual hazards separately. Fifth, the study sought to enhance community safety and preparedness by providing early warnings, thus reducing response times and improving the effectiveness of emergency responses. Finally, the system was thoroughly tested and evaluated under real-world conditions to ensure its reliability, accuracy, and effectiveness in delivering emergency notifications.

### **General Objectives:**

The general objective of this paper was to develop a monitoring system that made use of different sensing technologies in detecting potential hazards in homes such as fire, gas leaks, smoke, and earthquakes. The system was expected to alert residents quickly and correctly, thereby giving them a head start on reacting to such emergencies. The system applied sensors, notification platforms, and monitoring tools to quickly notify residents of a dangerous situation. This kept them safe, prevented damage to property, and reduced the risk of harm. It focused on making it easy for users, sending clear and timely notifications so that everyone in the household could respond appropriately in an emergency.

### **Specific Objectives:**

#### **1. Develop a device capable of detecting earthquakes, fire hazards, gas leaks, and smoke.**

Developed an automated safety device that integrated multi-sensor detection and machine learning to identify and respond to environmental hazards. The system utilized a Raspberry Pi Camera Module 3 with the YOLOv8n (You Only Look Once) algorithm for real-time fire detection and triggering an active alarm buzzer. An SW420 vibration sensor detected seismic activity, alerting users through a buzzer during repeated shaking events. Gas and smoke hazards were identified by an MQ2 Gas sensor, while a DHT22 sensor monitored temperature and humidity for risk assessment. All sensors interfaced with a Raspberry Pi 5, which processed data and relayed alerts to a mobile application. By combining AI-driven fire detection and multi-hazard monitoring, it enhanced early hazard response and reduced potential harm to occupants and property.

#### **2. Develop a device capable of notifying residents when earthquakes, fire hazards, gas leaks, and smoke are detected.**

Created a system capable of handling data from sensors for notifying residents about earthquakes, fire hazards, gas leaks, and smoke detection.

### **3. Develop a device that is capable of preventing fire in its early stage by using YOLOv8n and automated water.**

The device utilized a camera module connected to a Raspberry Pi 5 to continuously monitor its surroundings. It used the algorithms of YOLOv8n to detect the presence of flames or smoke in real-time. Upon fire detection, the system immediately triggered an audible alarm and activated connected water sprinklers. It simultaneously sent notification alerts to designated devices through a mobile application interface. The automated response mechanism provided fire suppression while maintaining optional manual control through physical switches or the application interface.

## **Scope and limitations of the study**

Researchers focused on the development and implementation of ISHA: Integrated Safety and Hazard Alert System for Home Emergencies. The system was designed to immediately notify the residents about the status of their homes and enable timely responses to unexpected emergencies. The main areas of research revolved around the deployment of IoT devices to ensure efficient data collection, processing and transmission of sensors, microcontrollers, and communication modules.

## **Scope**

The scope of ISHA: Integrated Safety and Hazard Alert System for Home Emergencies are the following:

## **Backend**

### **1.1 Admin Login Module**

Upon opening the application, two options popped up such as "Log in as

Admin" and "Continue as Guest." Choosing the "Continue as Guest" button, the user was redirected to the dashboard, and choosing "Log in as Admin" redirected the user to the login page where an admin passkey was required. After logging in, the user was redirected to the admin dashboard.

#### **1.1.1 Admin Camera Control**

This control helped the admin use the camera controls to freely pan and tilt the view. The Mist Button controlled the on or off status of the misting system.

#### **1.1.2 Incident Data Module**

In this module, the system maintained a record of past emergencies. The owner could view the list of past emergencies. Through the module, the system had a category and status for past emergencies.

### **Front End**

## **2.2 Dashboard**

### **2.2.1 Temperature and Humidity Monitoring Module**

The central display of the dashboard reads the current temperature and humidity of the place.

### **2.2.2 Gas Monitoring Module**

In this module, the application indicated "Normal" when the quality of air was safe, and it changed to a "Dangerous Gas Leak" alert once smoke, carbon monoxide, or methane were detected within the range of the sensor.

### **2.2.3 Vibration Monitoring Module**

In this module, the vibration rounds were indicated in the alert notifications

when consistent seismic activity was detected; the system displayed a red alert.

#### **2.2.4 Camera Live Feed**

This module helped users to access live feeds, ensuring accurate safety checks. The user had the ability to control the movement of the camera through a virtual joystick. When the camera detected a fire, it automatically located and highlighted the fire with a bounding box, and an automatic mist mechanism activated and put out the identified flame.

#### **2.2.5 Display Notification and Alerts Module**

This module provided real-time alerts and notifications to users regarding important updates and events. The module categorized three primary hazards such as Earthquake, Gas Leak, and Fire.

#### **2.2.6 Calling Emergency Hotline**

The emergency hotline section had the list of emergency hotlines based on the location of the users. To call a number, clicking the call button function redirected the users' phone to the phone's dialer to initiate the call. The "Add Emergency Contact" feature allowed users to customize and save their emergency contacts.

## **Limitations of the Study**

The limitation of ISHA: Integrated Safety and Hazard Alert System for Home Emergencies are the following:

### **1. Geographical Limitations**

The system will be tested in a limited area, which may affect the generalizability of the findings to larger or more diverse geographic areas.

### **2. Network Reliability**

The efficiency of a system is highly dependent on a stable Internet connection. System performance may suffer in areas with poor network infrastructure.

### **3. Emergencies**

The study mainly focused on general emergencies and did not include very complex or large-scale hazards (e.g., natural disasters, mass incidents).

## **Review of related literature/studies/systems**

This chapter covered the review of related research pertinent to the project. This discussed the similarities and differences between each study and drew insights on emergency response systems from these various works.

### **Local Literature**

According to INFORM, a global open-source risk assessment tool for humanitarian crises, the Philippines had a hazard exposure rate of 8.8, ranking it third among 191 assessed countries, indicating significant vulnerability to natural disasters. However, its coping capacity was only rated at 4.3, positioning it 104th globally, which highlighted the inadequate response capabilities of government institutions. Oganiza et al. (2021) emphasized that response time was critical in emergencies, urging local government units to develop emergency plans. This context was directly related to ISHA, which aimed to use IoT technology to improve real-time communication and response during disasters. By addressing the gaps in readiness and response highlighted by INFORM and Oganiza et al., ISHA strove to enhance resilience and effectively prevent the impacts of natural disasters.

As stated by Barredo et al. (2023), the researchers developed a system capable of detecting seismic activity and sending alerts to surrounding communities. The system used IoT technology to relay real-time data to authorities and the public, allowing them to prepare for an impending earthquake. While this research focused on large-scale disasters, its relevance to ISHA lay in its use of IoT to reduce response time. This similarly aimed to minimize delays in individual emergency situations by immediately notifying the residents to ensure that they were aware of any incident happening in their home.

According to Domingo et al. (2023), the demonstration of a device GSM and GPS modules was used to automatically send emergency alerts during accidents. This system was particularly notable for its use of technology to initiate automatic responses, which could potentially save lives. ISHA shared this characteristic but differentiated itself by integrating sensors to collect data and immediately notify the residents of any hazards.

In accordance with Barroquillo et al. (2024), exploring the creation of a communication system that utilized multiple methods such as SMS, mobile apps, and IoT devices guaranteed that emergency messages were sent even if one mode failed. The redundancy in communication methods ensured reliability in critical situations. This study provided a strong reference point for the ISHA system, which similarly emphasized the importance of ensuring that emergency alerts were sent and received without failure. By adding cameras for monitoring, ISHA also addressed the issue of false emergency alerts.

According to Batac et al. (2024), a portable device for individuals with disabilities, allowed them to send emergency alerts via SMS or phone calls. The device ensured that persons with disabilities could quickly access emergency services when in danger. Although ISHA was designed for the general population, this study was relevant as it demonstrated the importance of accessible emergency communication systems. ISHA also addressed accessibility but only within the house or the building.

### **Foreign Literature**

International research also supported the use of IoT in emergency response systems since it minimized the casualty by maximizing the travel time from the rescuers and victims. According to the study of Bacanin et al. (2023), the application of IoT in monitoring and managing disasters. The system used real-time data collection to send automatic alerts to authorities and nearby individuals when a disaster was detected. This system was highly relevant to this research model, as both relied on IoT technology to send alerts instantly, thereby reducing response time during emergencies. ISHA focused on notifying residents rather than responding to large-scale disasters, but the underlying technology principles were similar, with both systems aiming to save lives by providing fast and accurate information.

According to Chin Thomas (2023), researchers explore the role of local communities and governments in improving disaster preparedness and response. The study highlights the importance of collaboration between citizens and authorities in emergency situations, emphasizing that community-based systems are often more effective at managing crises. This study directly aligns with the goal of the ISHA system, which is designed to notify

residents and give information to contact local government units to ensure swift response to emergencies. The community-based approach was central to ISHA's mission of improving emergency communication and ensuring that every resident was aware.

As per Asyhari et al. (2021), exploring the latest advancements in IoT technology and its application in disaster monitoring was explained. The paper emphasized the role of sensors and communication networks in providing accurate, real-time data to emergency responders. Similar to ISHA, this system ensured that responders received detailed information about the emergency, allowing them to make informed decisions. While the focus here was on large-scale disasters, the core idea of using IoT to reduce response times and improve the accuracy of emergency reports was directly applicable to the ISHA system.

Industrial manufacturing processes had undergone a significant digital revolution during the past decade, resulting in more intelligent, adaptable, integrated, and efficient operations, a transformation known as 'Industry 4.0' or 'the fourth industrial revolution' Mojtabehi et al. (2021). This paradigm shift, initiated by a German government-funded paper in 2011, aimed at enhancing manufacturing competitiveness, encompassed increased digitization, adaptability, automation, virtualization, resource efficiency, and decentralization across various industries. By integrating technical techniques such as Artificial Intelligence (AI), Digital Twins, Blockchain, and Cloud computing, Industry 4.0 accelerated computerization and interconnection, leading to operational excellence. This framework was closely related to the ISHA, as both prioritized leveraging advanced technology to enhance responsiveness and efficiency. In emergency management, integrating IoT technology within ISHA aligned with Industry 4.0 goals by enabling real-time data collection and analysis critical for effective responses. The automation capabilities associated with Industry 4.0 further streamlined alert systems, facilitating immediate notifications to residents during hazardous kinds of situations. Ultimately, the connection between industrial advancements and community emergency response emphasized ISHA's potential to transform emergency systems, making them more efficient and effective in saving lives by harnessing modern technology.

Based on the previous study of Mohsin & Muyeved (2024), a smart emergency response system was designed to monitor vehicles, homes, and health statuses. This system used IoT to send real-time alerts to emergency services, ensuring that help arrived as quickly as possible. ISHA adopted a similar approach by using IoT to send alerts immediately when a hazard was detected. However, ISHA only functioned inside a home.

## **Synthesis**

Studies in the Philippine context emphasized the importance of timely responses and effective communication during emergencies. Oganiza et al. (2021) underscored the urgency of robust disaster response planning, a principle reflected in ISHA's design to enhance real-time communication. Barredo et al. (2023) and Domingo et al. (2023) showcased systems that leveraged IoT, GSM, and GPS to notify communities and authorities during seismic activity and accidents, highlighting similarities to ISHA's aim of reducing delays through automated notifications. Barroquillo et al. (2024) contributed insights on ensuring reliability through redundant communication methods, a feature ISHA incorporated alongside added monitoring capabilities to address false alerts. Batac et al. (2024) underscored accessibility for vulnerable populations, which, while designed for individuals with disabilities, informed ISHA's focus on making emergency alerts within homes effective and inclusive.

Global studies extended the discussion by showcasing IoT advancements in managing disasters. Bacanin et al. (2023) emphasized real-time data collection for automated disaster alerts, mirroring ISHA's real-time hazard notification system for homes. Chin Thomas (2023) highlighted the effectiveness of community-based systems, a principle that aligned with ISHA's collaboration with the locals to ensure an accurate response. Asyhari et al. (2021) demonstrated how IoT sensors and networks enhanced response accuracy, a key feature of ISHA's design. Similarly, Mojtahedi et al. (2021) connected the integration of advanced technologies under the Industry 4.0 paradigm to emergency management, suggesting that ISHA's use of IoT aligned with global trends of real-time automation and efficiency. Finally, Mohsin & Muyeved (2024) contributed insights into IoT-based emergency monitoring systems, similar on ISHA's approach to alerting residents.

## Hardware Descriptions

| Reference         | Type of sensors | Context of Use                                                                            | Description of Use                                                                                                                                                                                                             |
|-------------------|-----------------|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Barredo et al.    | IOT             | An Android Based Emergency Response Application with Web Based Emergency Report Generator | Used Android devices along with mobile Internet for communication and data transfer between users and emergency responders.                                                                                                    |
| Bacanin et al.    | IOT, AI         | Emergency Response and Disaster Management                                                | AI Integration, digitalization and fast action responses, real-time risk assessment, and detection for immediate alert                                                                                                         |
| Gatchalian et al. | IOT             | Government Text Alert System                                                              | Used application, wi-fi and mobile data for network provider, and technology adoption for direct message.                                                                                                                      |
| Mohsin et al.     | IOT, AI         | Smart emergency response systems for monitoring health, home and vehicle status           | Used Real-time monitoring, Gas sensor, flame sensor, humidity sensor, Temperature, and intrusion detection for ensuring the safety of the owner of the device and being up-to-date on the status of health, vehicle, and home. |
| Mojtahedi et al.  | IOT             | Industry 4.0, Disaster Risk Management and Infrastructure Resilience                      | Collecting data based on past research to improve the quality of future infrastructure and avoid any hazardous activity.                                                                                                       |
| Thomas, C.        | IOT             | Assessing the effectiveness of IoT in disaster management.                                | IoT was used to plan actions before, during, and after an emergency. It included training and trials to improve the effectiveness in emergency responses.                                                                      |
| Organiza et al.   | IOT             | Emergency Response Mobile Application                                                     | Real-time monitoring, faster response time, and tracker.                                                                                                                                                                       |

Table 1. Hardware Descriptions

## METHODOLOGY

### Overview of the Project

The project focused on developing an integrated safety observation system capable of detecting various domestic hazards such as fire, gas leaks, smoke, and earthquakes. The system sent real-time alerts to residents, enabling quick responses and enhancing home safety. It combined both hardware and software components, with sensors designed to detect specific hazards and a centralized software platform that processed the data and triggered appropriate notifications. The methodology involved extensive data collection through controlled experiments to evaluate the performance and accuracy of the sensors. Testing ensured the system was reliable, minimizing false alarms and optimizing response times. Additionally, user feedback was gathered to refine both the hardware and software, ensuring the system was user-friendly, effective, and ready for deployment in real-world scenarios.

### Prototype Model

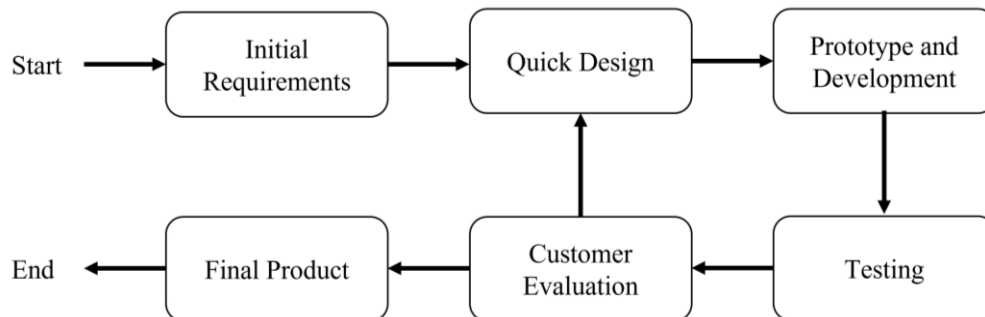


Figure 1. Prototype Model used in ISHA system

Figure 1. Prototype Model used in ISHA system to make the prototype of the project.

**Initial Requirements:** This procedure involved collecting and analyzing the basic requirements from researchers. Researchers engaged with users to understand their needs and expectations for the system. This part of the prototype model needed to provide project requirements along with user feedback.

**Quick Design:** Next, a quick design was created based on the gathered requirements. This quick design served as a blueprint for the prototype, providing a rough overview of how the system would function. It was not a complete design but rather a simplified version that satisfied in visualizing the end product.

**Prototype and Development:** The researcher built an initial prototype based on the quick design. A tangible representation of the concept was tested and evaluated. This prototype was a low-fidelity version of the final product, allowing users to interact with it and provide feedback.

**Testing:** After the prototype was developed, it was tested for adjustment. This phase involved checking system issues or errors regarding the prototype's functionality and usability. The researchers assessed strengths and weaknesses, which provided valuable insights into necessary improvements.

**Customer Evaluation:** This phase involved gathering user feedback from the evaluation phase. Researchers presented the system to users for evaluation. Researchers collected the issues or concerns raised by users. This process was continued until users were satisfied with the prototype.

**Final Product:** After refining the prototype and obtaining user's approval, researchers proceeded to create the final product. Next, involved implementing maintenance essential to ensure continued functionality. This phase included thorough testing of the system to ensure it met all requirements and functions as intended before deployment.

## **System Design Specification**

### **1. Framework Planning and Structural Design**

The Raspberry Pi 5 was the central microcontroller responsible for collecting data from various sensors and sending commands to the communication module. Each sensor played a vital role in detecting specific environmental data, such as gas leaks, temperature, smoke detection, and vibrations. The Android mobile application sent homeowners notifications about the facility's status when different sensors detected abnormality.

The system included mobile features that notified the residents even though they were not on the site. It was done by delivering a notification on the application that depended on the data detected by the sensors. The system of ISHA aimed to achieve reliable and scalable functionality, laying a solid foundation for future developments and system design enhancements.

### **2. Device Synchronization and Adjustment**

The sensors that the researchers used were MQ2 gas sensor, DHT22 temperature or humidity sensor, and a SW420 vibration sensor. Each sensor functioned depending on the range of their program; all sensors embedded in Raspberry Pi 5 were tested in different situations to determine the output accuracy and stored the data it gathered on the Android application.

The calibration took place in controlled environments for different testing procedures to determine the reliability of the device and to fix errors that were presented during the trials so it was ready to be put on different sites to ensure that the system could accurately produce data and the mobile application was capable of responding in case of emergency.

### **3. Application Programming and Engineering**

One aspect of software development focused on building Android mobile applications that had specific features. These features included real-time notification, enabling quick communication between the system and residents. The applications also had camera

features that could help to monitor the house. Additionally, homeowners could easily call for help even when they were not at home, by using the Emergency tab of the application as an emergency hotlines reference, enhancing overall safety and security.

For the Android mobile app development, a key feature was real-time communication. This meant that the applications could quickly notify occupants in case of emergencies. The apps were supported by different sensors, which allowed users to know the status of the situation based on the information received from the devices. The homeowners had the option of contacting responders even while they were away from their home, ensuring that help could arrive quickly when needed.

#### **4. On-Site Trials and Assessment**

The system was tested in an apartment with a minimum of 5 members in the family. The goals of testing the device in different situations were to test the response time, awareness it offered, and reliability in emergencies. In this session, participants were urged to ask questions and engage in discussions to improve their comprehension of how the system functioned.

Community training involved informative workshops for the landlord, teaching them how to engage with the system and what actions to take during an emergency. Offering practical experiences, like simulations or training scenarios, was helpful for the residents to feel more assured about utilizing the system. This cooperative method encouraged trust and collaboration, thereby improving the overall safety of the community, accuracy, and user satisfaction.

The Android mobile application received the detected emergencies from the sensors to notify the residents and to measure how quickly the system alerted via notification and how often it gave false alarms. User feedback was collected through surveys to understand how effective and easy to use both the system and Android mobile application were. The evaluation helped identify strengths and weaknesses, guiding improvements for better community safety.

## **5. Performance Review and Impact Analysis**

The data collected from field tests was used to evaluate how well the system operated in different situations. Improvements were made to the sensors' response times to ensure that both landlord and residents were quickly alerted in case an emergency occurred, which helped reduce damage in affected area.

To enhance security and decrease the chance of false alarms, applying a Raspberry Camera Module 3 helped with visualization on the system, to determine the status of the house. This design helped ensure that the residents were well protected by the system.

## **6. Deployment and Public Skill Enhancement**

A demonstration event was arranged to highlight the system's capabilities and the advantages it offered in emergencies. Making the residents proactive, it helped to reduce the chance of wide-scale accidents and damages on the property. Proper maintenance of the device was included in the demonstration for better performance and longer life span of the system.

## Conceptual Framework

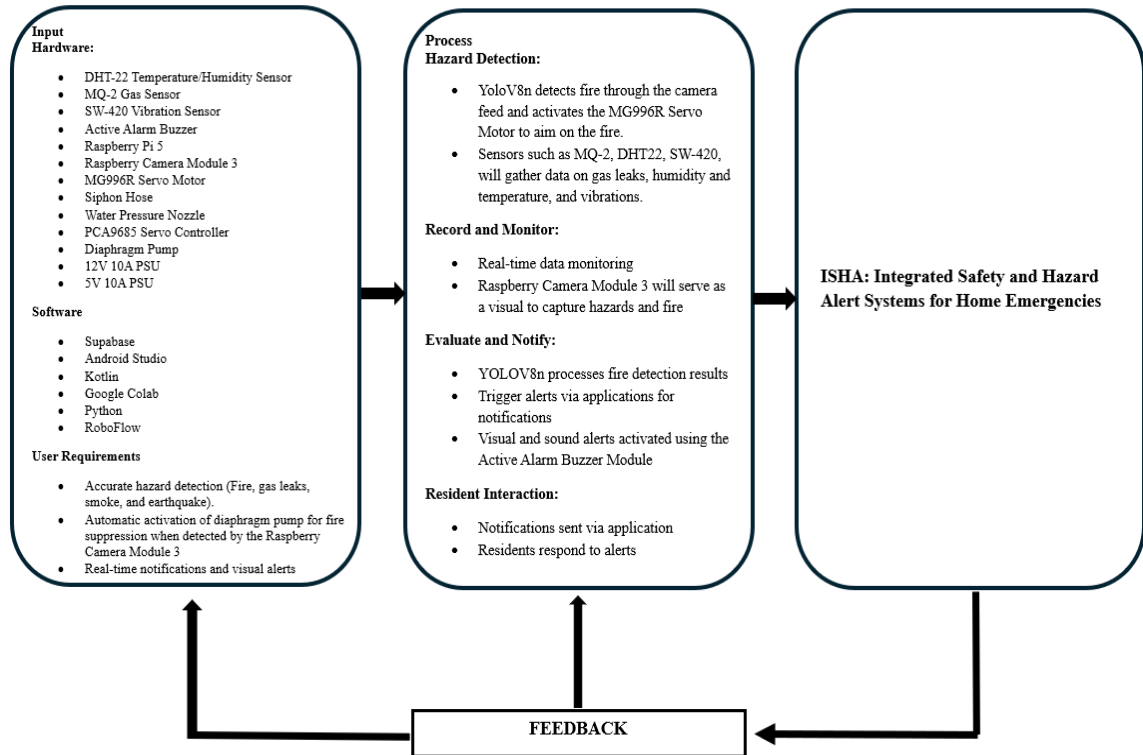


Figure 2. The Conceptual Framework of the ISHA system

The figure shows the elements and processes of the ISHA: Integrated Safety and Hazard Alert System for Home Emergencies, which are divided into three main phases: input, processing, and output. These components collectively enabled the system to meet its functional and operational requirements.

## System Block Diagram

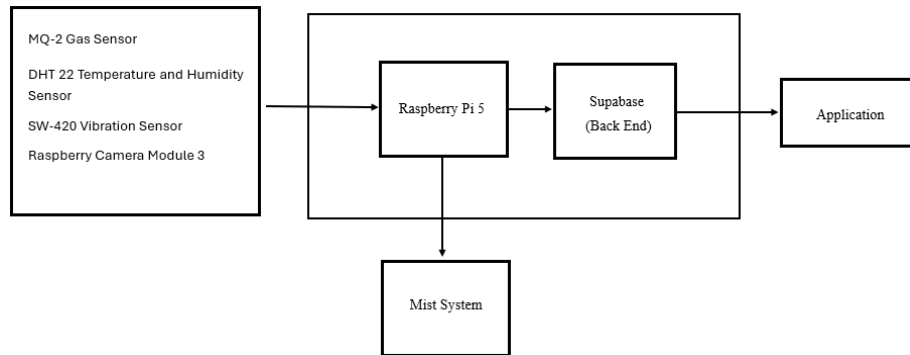


Figure 3. The System Block Diagram of ISHA system

The ISHA system's block diagram showcased how different components and subsystems were connected. Various sensors including the MQ-2 Gas Sensor, DHT22 Temperature/Humidity Sensor, and SW-420 Vibration Sensor gathered data. This data was then sent to the Raspberry Pi 5, which processed it and oversaw the system's operations. To keep residents informed about any potential hazards, the microcontroller used the Raspberry Camera Module 3 for visualization option when the occupants were in far distance and application for alerting the home owners and to send out timely and reliable alerts.

## Hardware/Software

### System Component (Hardware)

#### 1. Raspberry pi 5 (4Gb)

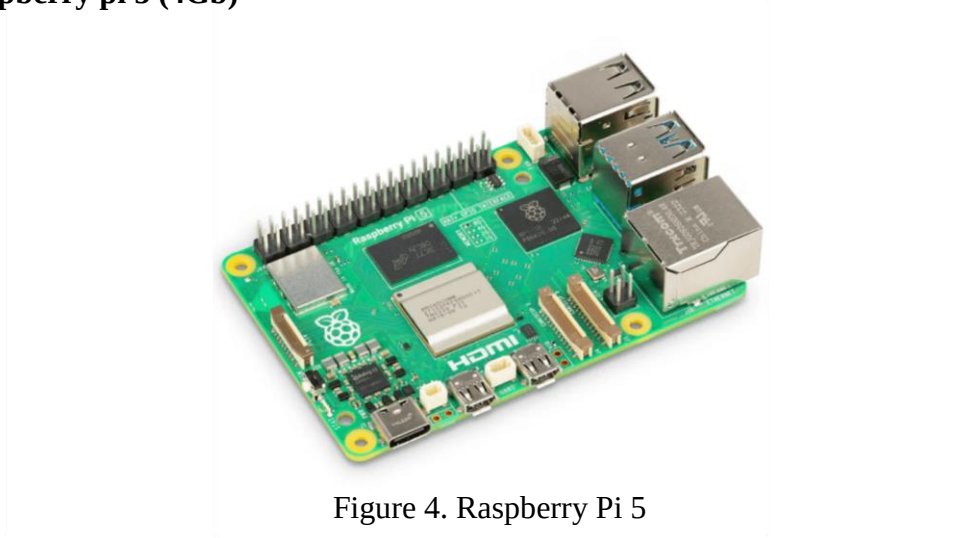


Figure 4. Raspberry Pi 5

The Raspberry pi 5 is a compact, low-power single-board computer designed for artificial intelligence (AI) and machine learning (ML) applications. It served as the main processing unit for YOLOv5-based fire detection.

#### 2. Raspberry pi aluminum armor case dual fan



Figure 5. Raspberry pi aluminum armor case dual fan

The Raspberry Pi aluminum armor case dual fan was used for controlling the temperature of Raspberry Pi 5 when various sensors like DHT22, MQ-2, and SW-420 were being used.

### 3. MQ-2 Gas Sensor



Figure 6. MQ-2 Gas Sensor

This device detected hazardous gases such as smoke, carbon monoxide, or methane. It helped identify potential fire or gas leak hazards, triggering alerts at dangerous level.

### 4. SW-420 vibration sensor

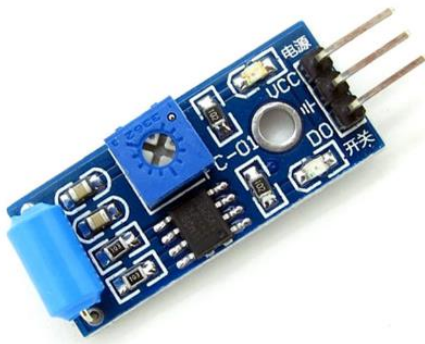


Figure 7. SW-420 vibration sensor

The SW-420 vibration sensor detected ground vibrations or seismic activity, alerting the system to the possibility of an earthquake or other seismic events.

## 5. DHT22 Temperature/Humidity Sensor

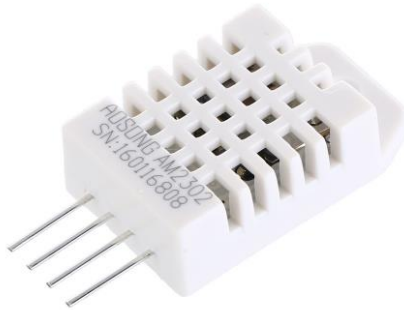


Figure 8. DHT22 Temperature/Humidity Sensor

This device was a temperature and humidity sensor that provided real-time data on environmental conditions, enabling the system to monitor temperature fluctuations that could indicate a fire or unsafe conditions.

## 6. Relay Module



Figure 9. Relay Module

The Relay Module was used by the Raspberry Pi 5 to control high-voltage hardware components such as a water pump and fans by isolating the control circuit from the power circuit.

## 7. Raspberry Pi Camera Module 3



Figure 10. Raspberry Pi Camera Module 3

The Raspberry Pi Camera Module 3 was a compact, high-resolution camera designed for embedded systems like the Raspberry Pi 5. It could capture clear and detailed images, making it suitable for hazard-monitoring applications. For this device, the camera module was used to document emergency events such as gas leaks, fires, or intrusions by capturing visuals of the affected area. These images provided critical context to help residents or responders assess the situation more effectively.

## 8. Active Alarm Buzzer Module

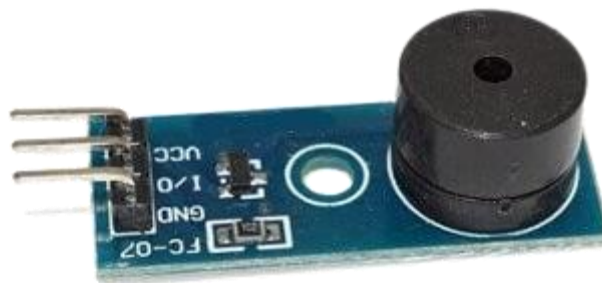


Figure 11. Active Alarm Buzzer Module

Buzzers were versatile audio signaling devices used in various applications, particularly in electronic projects and alarm systems. They were used to sound an alarm when hazards were detected.

## 9. MG996R Servos



Figure 12. MG996R Servos

The MG996R Servos are specialized electric motors designed for precise control of angular or linear position, speed, and acceleration. Servo motors were utilized in advanced mist systems, particularly for controlling the orientation and operation of spray nozzles.

## 10. Micro SD Card (32GB)



Figure 13. Micro SD Card (32GB)

For storing the operating system of Raspberry Pi 5 and libraries of YOLO algorithm for the camera.

## 11. 12V 10A Power Supply Unit (PSU)



Figure 14. 12V 10A Power Supply Unit (PSU)

A 12V 10A Power Supply Unit was a crucial component that was used to supply power to other hardware components to work, such as the water pump and fans.

## 12. Water Pressure Nozzle



Figure 15. Water Pressure Nozzle

This converted the water from the diaphragm water pump into pressurized water to mitigate fire.

### 13. Diaphragm Water Pump



Figure 16. Diaphragm Water Pump

The Diaphragm Water Pump generated pressurized water to hit the detected target of the camera module and to prevent fire hazards.

### 14. LM2596S Voltage Regulators



Figure 17. LM2596S Voltage Regulators

The Voltage Regulators supplied only the needed voltage of hardware components so that they could not be overloaded and ensured the safety of the components while being used.

### 15. PCA9685 Servo Controller



Figure 18. PCA9685 Servo Controller

It managed the precise movement of the servo motors, specifically 180 degrees. It allowed the main processor to control it for accurate positioning.

### 16. ¼ Silicone Hose



Figure 19. ¼ Silicone Hose

The Silicone Hose was connected to the diaphragm water pump and served as a tunnel for the water, going to the nozzle to extinguish fire.

### 17. 12V Fans



Figure 20. 12V Fans

The 12V Fans was used as a cooling device that maintained the temperature and avoided components overheating from long period of usage.

### 18. Siphon Hose



Figure 21. Siphon Hose

The Siphon Hose was connected to the servo pan tilt bracket, which was also connected to the Raspberry Pi Camera Module 3. This helped to automatically aim at the detected fire and served as a flexible passageway for the water coming from the diaphragm pump.

## 19. Servo Pan Tilt Bracket



Figure 22. Servo Pan Tilt Bracket

The Servo Pan Tilt Bracket was a mounting structure that allowed servo motors and the Rpi Camera Module to move vertically and horizontally up to 180 degrees to search and detect fire.

## System Components (Software)

### 1. Supabase



Figure 23. Supabase

Supabase Realtime and open-source Database was used to store data from the device's sensors, such as gas levels, temperature, humidity, and seismic activity. The mobile app fetched this data in real-time, allowing users to monitor conditions live. Additionally, Supabase did not require complex configuration, allowing developers to focus on building features such as gas leaks or seismic events, ensuring that users were promptly alerted. Supabase was highly cost-effective, with a free tier sufficient for initial deployments.

### 2. Android Studio

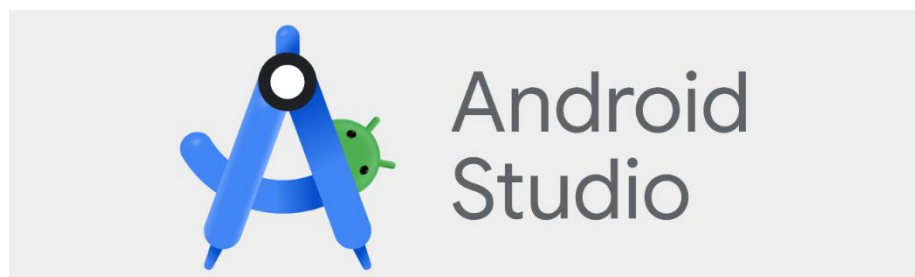


Figure 24. Android Studio

Android Studio was the official integrated development environment (IDE) designed for creating Android apps. It included specialized tools for advanced code editing, debugging, performance profiling, and visual interface design. This IDE enabled real-time testing on various virtual devices, which was critical for verifying that the emergency notification app

worked consistently across different Android platforms. By automating the development process and including the most recent Android SDKs, it assured a complete toolkit to make it easier to create high-quality, accessible mobile applications.

### 3. Kotlin



Figure 25. Kotlin

Kotlin was known as a modern, powerful programming language that Google had officially used for Android development. Its compact language removed unnecessary information, increasing developer productivity while promoting cleaner, more maintainable code. Kotlin's flawless compatibility with Java allowed for easy integration with existing libraries and tools, making it ideal for the project's Android application component. The language offered strong app performance across several platforms and included features such as null safety to reduce runtime errors. This made Kotlin ideal for creating dependable and efficient emergency alarm systems.

#### 4. Google Colab



Figure 26. Google Colab

Google Colab was used in the project as a cloud-based platform for developing, training, and testing machine learning models like the YOLOv8 fire detection algorithm. The cloud platform supported collaborative coding and version control, allowing the development team to effectively exchange and iterate on their work. Google Colab made it easier to experiment with and optimize AI models, which were important for reliable hazard identification in the system. The scalability and accessibility of Colab improved the team's ability to create reliable artificial intelligence solutions remotely.

#### 5. Python

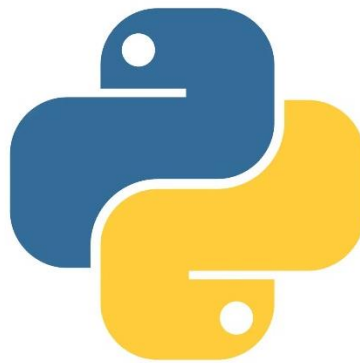


Figure 27. Python

Python was used for the machine learning aspect, enabling the real-time detection of hazards through the camera feed. It was used to program the Jetson Nano for processing fire detection through the YOLOv8 mode.

## 6. RoboFlow



Figure 28. RoboFlow

RoboFlow was used to manage and preprocess the image datasets required to train the project's fire detection models. It provided tools for data labeling, augmentation, and organizing, hence improving the quality and range of training data. This immediately improved the accuracy and generalizability of the YOLOv5 model when used for real-time hazard identification in the safety warning system. RoboFlow's user-friendly interface and integration features improved the dataset preparation workflow, promoting consistency and reducing manual errors. By optimizing the data flow using RoboFlow, the project ensured that the AI component was properly trained to respond to fire emergencies.

## System Overview

### View of ISHA

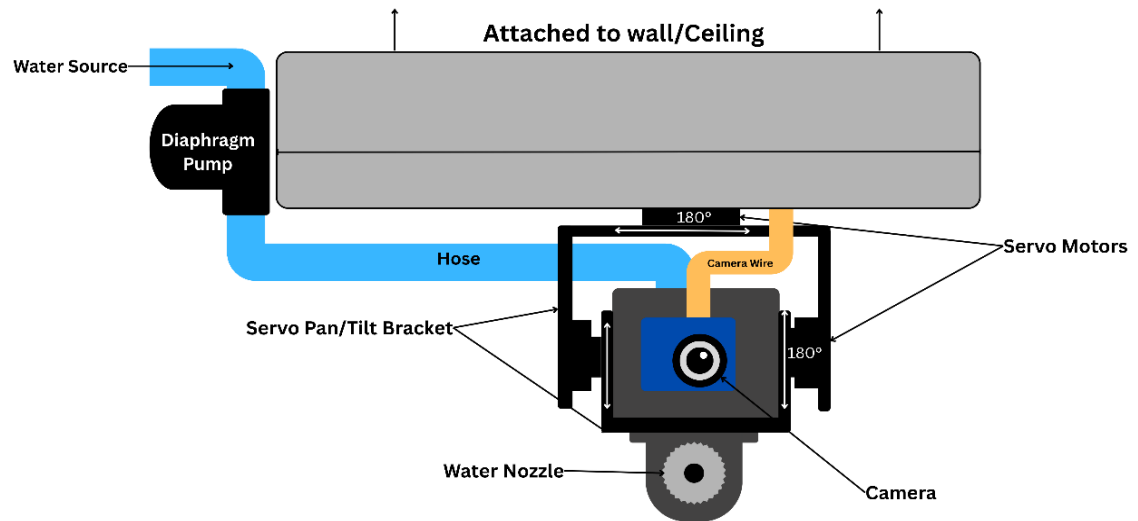


Figure 29. View of ISHA

Figure 29 presents the ISHA system overview of the hardware of the project created using Canva.

## Wireframing

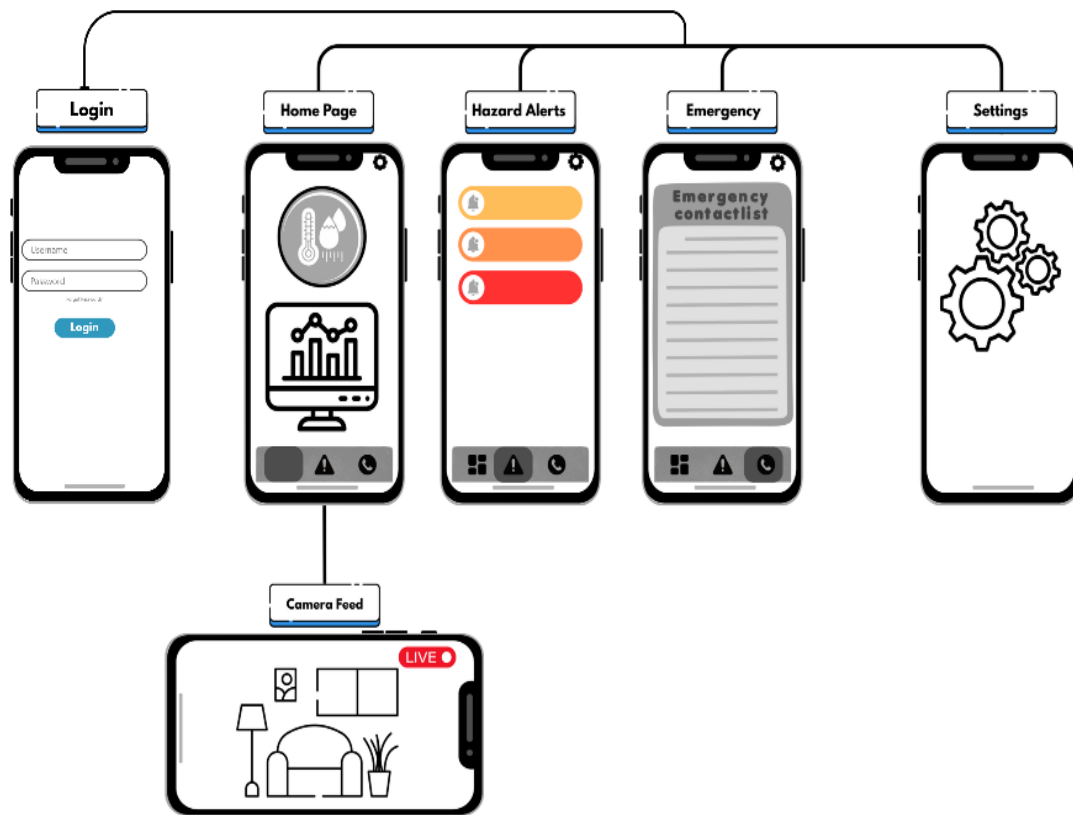


Figure 30. Wireframing

## Calendar of Activities

[illegible]

Table 2: Gantt Chart of Activities

Table 2 represented the calendar activities that listed the important tasks and their due dates from September 2024 to November 2025 and provided a thorough timeline for the paper process. It covered initial actions like planning, adviser consultations, and presenting and approving the suggested device. To ensure timely progress, the chart also followed the creation and revision of the study title, Chapter I, II, III, IV and V. It also highlighted specific events, such as weekly target goals and submissions. This well-planned timetable guaranteed that all objectives were reached within the allotted time and helped keep focus.

**Budgetary Estimate**

| Quantity      | Specifics                                 | Approximate Cost (Philippine Peso) |
|---------------|-------------------------------------------|------------------------------------|
| 1             | Raspberry pi 5 4gb                        | ₱6,499                             |
| 1             | Raspberry pi aluminum armor case dual fan | ₱500                               |
| 1             | MQ-2 Gas Sensor                           | ₱75                                |
| 1             | SW-420 Vibration Sensor                   | ₱70                                |
| 1             | DHT22 Temperature/Humidity Sensor         | ₱125                               |
| 1             | Relay Module                              | ₱150                               |
| 1             | Raspberry pi Camera Module 3              | ₱2,190                             |
| 1             | Active Alarm Buzzer                       | ₱29                                |
| 1             | Mg996r Servo Motor                        | ₱250                               |
| 1             | SD Card (32gb)                            | ₱350                               |
| 1             | 12V 10A Power Supply Unit (PSU)           | ₱450                               |
| 1             | Water Pressure Nozzle                     | ₱180                               |
| 1             | Diaphragm Water Pump                      | ₱400                               |
| 1             | LM2596s Voltage Regulator                 | ₱45                                |
| 1             | XL4015 Voltage Regulator                  | ₱97                                |
| 1             | Pca9685 Servo Controller                  | ₱250                               |
| 1             | ¼ Silicone Hose                           | ₱110                               |
| 2             | 12V Fan                                   | ₱115                               |
| 1             | Siphon Hose                               | ₱90                                |
| 1             | Servo Pan Tilt Bracket                    | ₱250                               |
| <b>Total:</b> |                                           | <b>₱12,225</b>                     |

Table 3. Budgetary Estimate of the ISHA machine

Table 3 represents the Budgetary Estimate table of the total expenses of the researchers for each hardware component.

## Human Resources

The following pages contain the curriculum vitae of the researchers and the Adviser's Acceptance Form.

Curriculum Vitae of  
**JOHN ALLAIN P. ACCAD**  
**Alimasag St. Barangay 12 Caloocan City**  
**accad.309444@caloocan.sti.edu.ph**  
**09613890838**



### EDUCATIONAL BACKGROUND

| Level                | Inclusive Dates | Name of school/ Institution     |
|----------------------|-----------------|---------------------------------|
| Tertiary             | 2022 – Present  | STI College Caloocan            |
| Vocational/Technical | 2020 – 2022     | University of the East Caloocan |
| High School          | 2016 – 2020     | Longos National High School     |
| Elementary           | 2009 – 2016     | Ninoy Aquino Elementary School  |

### PROFESSIONAL VOLUNTEERING EXPERIENCE

| Inclusive Dates | Nature of Experience/<br>Job Title | Name and Address of Company or<br>Organization |
|-----------------|------------------------------------|------------------------------------------------|
| 2015-2016       | Volunteer                          | Batang Bert Emergency Response<br>Team         |

### AFFILIATIONS

| Inclusive Dates | Name of Organization                     | Position |
|-----------------|------------------------------------------|----------|
| 2020-2022       | University of the East: Computer<br>Club | Member   |
| 2019-2020       | LNHS: Pinsel Club                        | Editor   |

### SKILLS

| SKILLS            | Level of Competency | Date Acquired |
|-------------------|---------------------|---------------|
| Computer Literate | Advance Beginner    | 2022          |
| Java Fundamentals | Advance Beginner    | 2023          |

### TRAININGS, SEMINARS OR WORKSHOP ATTENDED

| Inclusive Dates | Title of Training, Seminar or Workshop |
|-----------------|----------------------------------------|
| June 2023       | NSTP Training                          |
| May 2024        | National Youth Convention              |

Curriculum Vitae of  
**RHEN L. BERDOL**  
**114G Alpine Alley St. Brgy 151 Caloocan City**  
**berdol.309998@caloocan.sti.edu.ph**  
**09941269927**



#### EDUCATIONAL BACKGROUND

| Level                | Inclusive Dates | Name of school/ Institution     |
|----------------------|-----------------|---------------------------------|
| Tertiary             | 2022 – Present  | STI College Caloocan            |
| Vocational/Technical | 2020 – 2022     | San Francisco High School       |
| High School          | 2016 – 2020     | San Francisco High School       |
| Elementary           | 2009 – 2016     | Bagong Barrio Elementary School |

#### PROFESSIONAL VOLUNTEERING EXPERIENCE

| Inclusive Dates | Nature of Experience/<br>Job Title | Name and Address of Company or<br>Organization |
|-----------------|------------------------------------|------------------------------------------------|
|-----------------|------------------------------------|------------------------------------------------|

#### AFFILIATIONS

| Inclusive Dates | Name of Organization | Position           |
|-----------------|----------------------|--------------------|
| 2024-2025       | Honor Society        | CPE Representative |

#### SKILLS

| SKILLS            | Level of Competency | Date Acquired |
|-------------------|---------------------|---------------|
| Computer Literate | Advance Beginner    | 2022          |
| Java Fundamentals | Advance Beginner    | 2023          |

#### TRAININGS, SEMINARS OR WORKSHOP ATTENDED

| Inclusive Dates | Title of Training, Seminar or Workshop |
|-----------------|----------------------------------------|
| June 2023       | NSTP Training                          |
| May 2024        | National Youth Convention              |

Curriculum Vitae of  
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**116-A Pineapple Rd. Potrero Malabon City**  
**narrciso.317728@caloocan.sti.edu.ph**  
**09945620067**



#### EDUCATIONAL BACKGROUND

| Level                | Inclusive Dates | Name of school/ Institution        |
|----------------------|-----------------|------------------------------------|
| Tertiary             | 2022 – Present  | STI College Caloocan               |
| Vocational/Technical | 2017 – 2019     | Arellano University Elisa Esguerra |
| High School          | 2013 – 2017     | Potrero National High School       |
| Elementary           | 2007 – 2013     | Potrero National High School       |

#### PROFESSIONAL VOLUNTEERING EXPERIENCE

| Inclusive Dates | Nature of Experience/<br>Job Title | Name and Address of Company or<br>Organization |
|-----------------|------------------------------------|------------------------------------------------|
|-----------------|------------------------------------|------------------------------------------------|

#### AFFILIATIONS

| Inclusive Dates | Name of Organization | Position |
|-----------------|----------------------|----------|
|-----------------|----------------------|----------|

#### SKILLS

| SKILLS            | Level of Competency | Date Acquired |
|-------------------|---------------------|---------------|
| Computer Literate | Advance Beginner    | 2022          |
| Java Fundamentals | Advance Beginner    | 2023          |

#### TRAININGS, SEMINARS OR WORKSHOP ATTENDED

| Inclusive Dates | Title of Training, Seminar or Workshop |
|-----------------|----------------------------------------|
| June 2023       | NSTP Training                          |
| May 2024        | National Youth Convention              |

Curriculum Vitae of  
**KARL EDISON R. NATIVIDAD**  
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**natividad.185040@caloocan.sti.edu.ph**  
**09297010759**



#### EDUCATIONAL BACKGROUND

| Level                | Inclusive Dates | Name of school/ Institution |
|----------------------|-----------------|-----------------------------|
| Tertiary             | 2022 – Present  | STI College Caloocan        |
| Vocational/Technical | 2019 – 2021     | STI College Caloocan        |
| High School          | 2015 – 2019     | Obando National High School |
| Elementary           | 2009 – 2015     | Binuangan Elementary School |

#### PROFESSIONAL VOLUNTEERING EXPERIENCE

| Inclusive Dates | Nature of Experience/<br>Job Title | Name and Address of Company or<br>Organization |
|-----------------|------------------------------------|------------------------------------------------|
|-----------------|------------------------------------|------------------------------------------------|

#### AFFILIATIONS

| Inclusive Dates | Name of Organization | Position |
|-----------------|----------------------|----------|
|-----------------|----------------------|----------|

#### SKILLS

| SKILLS            | Level of Competency | Date Acquired |
|-------------------|---------------------|---------------|
| Computer Literate | Advance Beginner    | 2021          |
| Java Fundamentals | Advance Beginner    | 2022          |

#### TRAININGS, SEMINARS OR WORKSHOP ATTENDED

| Inclusive Dates | Title of Training, Seminar or Workshop |
|-----------------|----------------------------------------|
| June 2021       | NSTP Training                          |
| May 2023        | Packetworx Internet of Things          |

Curriculum Vitae of  
**RUZZEL P. TEOTICO**  
**349 Katipunan Street 2nd Avenue Caloocan City**  
**teotico.233553@caloocan.sti.edu.ph**  
**09859194727**



#### EDUCATIONAL BACKGROUND

| Level                | Inclusive Dates | Name of school/ Institution     |
|----------------------|-----------------|---------------------------------|
| Tertiary             | 2022 – Present  | STI College Caloocan            |
| Vocational/Technical | 2020 – 2022     | STI College Caloocan            |
| High School          | 2016 – 2020     | F.G Calderon Integrated School  |
| Elementary           | 2009 – 2016     | Barrio Obrero Elementary School |

#### PROFESSIONAL VOLUNTEERING EXPERIENCE

| Inclusive Dates | Nature of Experience/<br>Job Title | Name and Address of Company or<br>Organization |
|-----------------|------------------------------------|------------------------------------------------|
|-----------------|------------------------------------|------------------------------------------------|

#### AFFILIATIONS

| Inclusive Dates | Name of Organization | Position |
|-----------------|----------------------|----------|
|-----------------|----------------------|----------|

#### SKILLS

| SKILLS            | Level of Competency | Date Acquired |
|-------------------|---------------------|---------------|
| Computer Literate | Advance Beginner    | 2022          |
| Java Fundamentals | Advance Beginner    | 2023          |

#### TRAININGS, SEMINARS OR WORKSHOP ATTENDED

| Inclusive Dates | Title of Training, Seminar or Workshop |
|-----------------|----------------------------------------|
| June 2023       | NSTP Training                          |
| May 2024        | National Youth Convention              |

## **ADVISER'S ACCEPTANCE FORM**

**NAME OF PROPONENTS:**

John Allain P. Accad  
Rhen L. Berdol  
Jerome F. Narciso  
Karl Edison R. Natividad  
Ruzzel P. Teotico

**APPROVED RESEARCH TITLE:**

ISHA: Integrated Safety and Hazard Alert  
System for Home Emergencies

**AREA OF STUDY:**

Internet of Things (IoT)

**CONFORME:**

Mr. Cyber Celwin Popanes  
**Capstone Project Adviser**

**APPROVED BY:**

Engr. Jimmy J. Jovero  
**Capstone Project Coordinator**

**Date: November 2025**

**NOTED BY:**

Mr. Winchester Pastolero  
**Program Head**

## RESULTS AND DISCUSSIONS

This section detailed the specific role and observed performance of each major component in realizing the system's core objectives: detection, alerting, mitigation, and monitoring. The results of the hardware implementation confirmed the functionality and integration of the selected components in creating the Integrated Safety and Hazard Alert System for Home Emergencies (ISHA).

### HARDWARE

#### Raspberry Pi 5

To start, Raspberry Pi 5 was the central processing unit, connected to all peripherals and sensors and actuators via General-Purpose Input/Output (GPIO) pins and communication protocols. The system's emergency alert logic and object detection routine (YOLOv8) were executed on this device, supplying sufficient power and connectivity to manage simultaneous data streams from multiple sensors. Its selection was justified by its ability to handle complex computational tasks, such as real-time video analysis and synchronous collecting of data from multiple sensors, which was critical for an effective emergency system.

#### Results

- |                                           |                                                               |
|-------------------------------------------|---------------------------------------------------------------|
| ✓ Successfully connected all the sensors. | ✗ Using peripherals, sensors, and all servo                   |
| ✓ The information was in real-time basis. | motors simultaneously were not possible due to lack of power. |
| ✓ All functions are usable.               | ✗ Was prone to overheating for long periods of time.          |

## Discussions

The results were successful in using Raspberry Pi 5 to be used on tasks that required rapid delivering of information. All the sensors were working and the data that it gave was being monitored by the user with the use of a screen or monitor. However, lack of supply occurred when peripherals were being added on the system while all the functions of the sensors and servo motors were being used, it could be improved by upgrading to a much larger voltage and ampere power supply unit.

### MQ-2 Gas Sensor

The early detection of flammable gas and smoke was a vital function of the sensor to identify leaks from the house. Upon detecting smoke and gas such as LPG, Alcohol, propane, Hydrogen, Carbon monoxide, and methane, the sensors promptly transmitted signals to the Central Processing Unit to trigger necessary warning protocols.

## Results

- |                                                               |                                                            |
|---------------------------------------------------------------|------------------------------------------------------------|
| ✓ Successfully triggered alerts upon detecting gas and smoke. | ✗ Covers only a limited distance, approximately 90–100 cm. |
| ✓ A minimal delay on detecting at long distance.              | ✗ Can be too small for large environment.                  |

## Discussions

The testing is successful in detecting of concentrations of gas above the level of threshold that has been set. The reliability of the sensor generates necessary digital signal output,

which was interpreted by the Raspberry Pi 5. The performance of MQ-2 Gas Sensor is suited for detecting the air quality, smoke, and possible leaks on indoor environment.

### **SW-420 Vibration Sensor**

The SW-420 Vibration Sensor is responsible for detecting vibrational patterns that can indicate for physical events such as earthquake. This will serve as both monitoring and trigger of earthquake, differentiating a normal house vibration to a potentially dangerous vibration. Monitoring the intensity and duration of the movement can provide a reliable input for issuing alerts during critical ground disturbances.

### **Results**

- |                                                                    |                                                                                         |
|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| ✓ Successfully triggered alerts upon detecting intense vibrations. | ✗ Too much intensity can damage the sensor physically.                                  |
| ✓ A buzzer will be triggered.                                      | ✗ A dedicated place for the sensor is necessary to prevent receiving false information. |
| ✓ Data can be seen on the screen.                                  |                                                                                         |

### **Discussions**

The results can be differentiated from high and low when it detects a vibration, the systems alerts are successfully triggered during the testing, ensuring the reliability of the sensors on the role of detecting vibration in critical moments and triggering the functionality of the emergency alert system.

## **DHT-22 Temperature and Humidity Sensor**

Using DHT-22 Sensor is utilized as a form of monitoring instrument to acquire accurate readings of the immediate change on temperature and humidity levels. The process of data is continuous since this is a crucial part for detecting a sudden rise on normal room temperature. A rising of temperature in the room could signify a pre-fire condition requiring the data analysis of the sensor before it triggers the alert system.

### **Results**

- |                                                                      |                                                                                       |
|----------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| ✓ A slight change on temperature and humidity is detected.           | ✗ Receive a maximum reading of 80° C.                                                 |
| ✓ Alert will only trigger when the temperature is constantly rising. | ✗ The readings can be change manually by using lighters and other fire related tools. |

### **Discussions**

The results provide a highly consistent and stable environmental readings across multiple test cycles. The data confirmed that the sensors output accurately reflected on both minor and major changes in temperature and relative humidity. The stability and precision of the data output make the sensor a reliable source of information for both normal room temperature and a sudden rise of temperature.

## **Raspberry Pi Cam Module 3**

The camera module provides visual feed for the fire detection model. The use of Raspberry Pi 5 allowed the camera module 3 to monitor video in real-time and accurately identifying fire instances within its field of view.

## Results

- ✓ A little delay on camera movement occurs when it is controlled on application.
- ✓ A clear visual for the user.
- ✓ Can monitor different angle and positions.
- ✗ Low Fps.
- ✗ Colors may vary depending on the lighting in the location.

## Discussions

The results of implementing RPI Cam Module 3 were successful since it provided necessary high-resolution visual input to the Raspberry Pi 5 for continuous monitoring of the indoor home. The image had a minimal delay while performing real-time fire detection. The RPI Cam Module 3 was an effective and reliable component for the visual-based detection of the user.

### **PCA 9685, MG996R Servo Motor, and Servo Pan Tilt Bracket**

The mechanical assembly of the Servo Pan Tilt Bracket provided the mounting that was necessary for high-precision aiming functionality. The PCA9685 Servo Controller controlled the precision and 180° movement of the MG996R Servo Motor on both vertical and horizontal axes, demonstrating the usefulness of the I2C protocol in overcoming the Raspberry Pi's resource restrictions. This controlled movement allowed the camera and the associated siphon hose to be swiftly and accurately directed to the coordinates of the fire detected by the RPI Camera Module 3.

## Results

- ✓ Responsiveness of MG996R Servo motor movement when being controlled in the application was observed.
- ✓ Reliability of Servo Pan Tilt Bracket that served as a mechanical mount for the camera, hose, and servo motor was established.
- ✓ Stopped and Focused on the place when the RPI Camera Module 3 detects fire while surveying the area.
- ✓ Easy to set up.
- ✗ Tangled wire can cause trembling of the servo motor.
- ✗ Sometimes a slight delay occurred.

## Discussions

The Servo motor movements were successful, and it focused on the fire when it was detected by the camera. The performance of remotely controlling it could help, especially when the user was far from the device, to check the status of the house once the alarm had been triggered.

### Power Supply Unit (PSU)

The PSU was responsible for converting the main power into specific, regulated voltages and currents that were needed by every component in the system. It ensured that components received stable and enough power to perform the specific tasks of each component. The PSU prevented system instability and protected electronics from damage due to overload or under voltage conditions.

## Results

- |                                                                              |                                                                                      |
|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| ✓ Supply sufficient voltage for each component.                              | ✗ A lack of voltage when peripherals are added while all the components are running. |
| ✓ Different PSU for servo motors, fans, and sensor to prevent under-voltage. |                                                                                      |

## Discussions

The PSU provided regulated power across all required outputs; thus, the test was successful. The consistent voltage levels under different conditions during the operation of the pump, fans, and CPU meant it delivered reliability in preventing component damage and ensured that even when the components required different voltage, the PSU met the power demands of the components without causing fluctuations.

### Active Alarm Buzzer

The Active Alarm Buzzer served as an audible warning device to immediately notify the occupants when a hazard was detected, such as gas leaks, vibrations, or fire. Its function was to convert alert signals from the Raspberry Pi 5 into a loud sound in critical scenarios.

## Results

- |                                                                               |                                                |
|-------------------------------------------------------------------------------|------------------------------------------------|
| ✓ The sound was loud that could be heard across the house.                    | ✗ Sometimes the buzzer fails to produce sound. |
| ✓ The sound could immediately catch the attention of anyone within its range. | ✗ Was easy to break.                           |

## Discussions

The buzzer was an effective component to use in alert systems because it provided a high-decibel signal immediately when any hazards such as gas, vibration, or fire were detected. Its high-pitch sound and fast activation were crucial for early warning signs since information needed to be communicated instantly without relying on network connectivity.

### 12V Exhaust Fans

The 12V Exhaust Fans were designed to actively regulate the temperature within the enclosed housing of the hardware components. Continuous draw of cool air into the device and removing the hot air generated by the Raspberry Pi 5 and PSU was critical for avoiding overheating when being used for a long period of time.

## Results

- |                                             |                                                         |
|---------------------------------------------|---------------------------------------------------------|
| ✓ Effective components against overheating. | ✗ Requires different power supply to operate correctly. |
| ✓ Regulated the air flow.                   |                                                         |
| ✓ Prevents the breakdown of devices.        |                                                         |

## Discussions

The 12V Exhaust Fans maintained a consistent airflow, regulating the internal temperature of the closed enclosure successfully even during long periods of workload. The performance of an exhaust fan validated the thermal management of other components and ensured the longevity of component life and reliability of the device.

## **SOFTWARE**

### **Mobile Application (Android) Development**

ISHA's mobile application was built using Android Studio with Kotlin as the primary programming language. It offered an easy and engaging interface for monitoring real-time sensor data, receiving warning notifications, and seeing live camera feeds. The application connected to the Supabase PostgreSQL database, allowing for instant changes from the Raspberry Pi. Compatibility testing was carried out on various Android devices to ensure smooth performance.

### **Results**

- |                                                                                                  |                                                                    |
|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| ✓ Displays real-time sensor data and camera feed.                                                | ✗ Notification delay during weak internet connection was observed. |
| ✓ User interface was responsive and Notification delay occurred during weak internet connection. |                                                                    |
| ✓ Stable synchronization with Supabase was achieved.                                             |                                                                    |

### **Discussions**

The ISHA mobile application performed effectively during testing, providing users with a clean and responsive interface for monitoring security risks. Under normal situations, real-time updates and notifications worked properly. Although there was a small delay in notifications due to bad connectivity, it did not hinder the system's general usability. The mobile application improved the system's responsiveness and accessibility, allowing users to respond fast in situations.

## Back-End Database Structuring

The Supabase PostgreSQL database served as the primary storage and communication hub for the Raspberry Pi and the mobile application. It saved structured sensor data, dates and times, and alarm information in SQL databases and enabled effective real-time synchronization over connected devices using Supabase's Realtime API.

## Results

- |                                                                                   |                                                        |
|-----------------------------------------------------------------------------------|--------------------------------------------------------|
| ✓ Stored and updated multiple sensor data efficiently in a structured SQL schema. | ✗ Minor delay during Simultaneous multi-sensor updates |
| ✓ Maintained real-time synchronization with the Raspberry Pi.                     |                                                        |
| ✓ Ensured data integrity and reliability with relational constraints.             |                                                        |

## Discussions

The database effectively managed all sensor data in a structured SQL context while also ensuring hardware and mobile interface synchronization. The small delay caused by many simultaneous updates was minimal and expected due to network latency. Overall, Supabase provided a solid and reliable data platform that enabled real-time monitoring and communication, enhancing the ISHA system's dependability.

## YOLOv8 Fire Detection Software Model

The YOLOv8 object detection model was used on the Raspberry Pi to detect fire via the camera stream. The model was trained on datasets processed using RoboFlow and Google Colab before being combined with Python and OpenCV for real-time detection.

## Results

- ✓ Detects visible fire accurately within 1–2 seconds
- ✗ False positives in dim lighting conditions
- ✓ Automatically activates mist system upon detection
- ✓ High model accuracy and fast image processing

## Discussions

The YOLOv8 model showed efficient detection and classification of fire hazards during testing. Its quick response time ensures that the mist system is activated and alert notifications are received on time. However, occasional false detections were seen in low-light circumstances, indicating the need for dataset augmentation or illumination modifications. Despite this, AI integration significantly increased ISHA's detection capability and system intelligence.

## Supabase Realtime and Alert System

The Supabase Realtime API handled alert notifications, sending hazard alerts to the user's mobile device whenever sensor readings exceeded predefined criteria by monitoring database changes.

## Results

- ✓ Sends real-time notifications to multiple devices via database listeners.
- ✗ Notification delay occurred under poor network connectivity.
- ✓ Provided clear and detailed alert messages.
- ✓ Consistent synchronization with database alerts.

## Discussions

The alert system effectively supplied precise and on-time notifications, keeping users informed during emergency situations. Including details such as hazard type and location increased user awareness and responsiveness. Despite delays caused by unreliable connections, all alerts were eventually received. The system's dependable alert transmission boosted ISHA's position as a responsive home safety solution.

### **Web Server and Cloud Integration**

The Raspberry Pi's built-in Wi-Fi module allowed connection between the system and the Supabase cloud. This enabled continuous data transfer, remote access, and real-time monitoring via the mobile app.

### **Results**

- |                                                         |                                                        |
|---------------------------------------------------------|--------------------------------------------------------|
| ✓ Maintained stable cloud connection for data exchange. | ✗ Minor latency occurred during heavy network traffic. |
| ✓ Allowed remote monitoring from mobile application.    |                                                        |
| ✓ No data loss occurred during transmission.            |                                                        |

### **Discussions**

All software components were able to communicate smoothly and reliably because of the cloud integration's effective performance. No data was lost or corrupted amid bandwidth bottlenecks despite a little lag. This demonstrated the system's cloud infrastructure's reliability and capability for ongoing operation in practical settings. These outcomes confirmed the ISHA system's effectiveness as a reliable and responsive home emergency monitoring system.

## CONCLUSION

According to the survey, respondents from age 21 to 30 years old showed higher familiarity with smart systems and monitoring devices than any of the age brackets. This familiarity enabled individuals to quickly realize the device's operation and easily connect their point of view with the researchers' plan for the system. The ages 31 to 40 years old were a little bit hesitant about the reliability and safety of the device; respondents in this age bracket were also concerned about the electricity bill that would cause the system since it would serve as a monitoring device for their homes. From the Ages 41 and above group, they were composed of workers, retired, and grandparents. They were open-minded about the system but they were not familiar with the device and its components. At least 2 out of 5 respondents from this age bracket wanted to stay on the traditional way of having safety equipment in their house rather than having technology and depended on it, in case of emergency.

This study successfully created and implemented the Integrated Safety and Hazard Alert System for Home Emergencies (ISHA), which combined multiple hardware and software components to meet four primary safety objectives: detection, alerting, mitigation, and monitoring. The system's performance was demonstrated by careful testing of its central processing unit, multiple sensor network, mechanical action components, and user-facing mobile application.

The study's findings supported the good performance of the combined components. The Raspberry Pi 5 successfully served as the central processing unit, carrying out complicated tasks like the real-time YOLOv8 object detection algorithm and managing multiple data streams from all sensors. Major detection components, such as the MQ-2 Gas Sensor, SW-420 Vibration Sensor, and DHT-22 Temperature and Humidity Sensor, consistently and accurately identified critical hazards such as gas leaks, significant vibrations, and sudden temperature increases, triggering the appropriate alert protocols.

The mechanical subsystem, which included the PCA9685, MG996R Servo Motor, and Servo Pan Tilt Bracket, successfully delivered the high-precision, remotely controlled movement needed for camera orientation and potential fire prevention. To ensure user accessibility, the ISHA mobile application and Supabase PostgreSQL database ensured real-time data synchronization and offered an intuitive interface for remote monitoring and instant hazard notifications.

While minor limitations were noted, such as limited power supply capacity when running all peripherals at the same time and slight notification delays under poor network conditions, overall testing confirmed the system's ability to function as an effective and reliable emergency safety solution for indoor environments. The research demonstrated the successful integration of AI-based detection and smart alarm systems on a tiny platform to improve home safety.

## RECOMMENDATIONS

Like any system, the ISHA alert system for home emergencies has its flaws, and there is potential for enhancement. The suggestions listed below are intended for future development.

**Improving the security of the system.** Putting a strong system security is highly recommended. It can protect the users from any kind of unauthorized access and data breaches. By adding stronger authentication, it will keep the system safe and reliable during emergencies.

**Implementation of backup power supply.** Adding a backup power supply is recommended because it can keep the system active in case outages occur. This will prevent delays in monitoring and will ensure continuous protection even when there is no power available.

**Adding backup communication options.** This is very important because if the main network fails, the residents will still receive the alerts coming from the device. Bluetooth or offline alarms are good options that can help to maintain reliable notifications in critical conditions.

**Using More durable hose.** To guarantee that the mist system operates safely and reliably when the device detects fire, it is recommended to use a more durable hose. In real fire situations, a stronger hose increases system dependability by preventing bursting, leaks, and pressure loss.

**Integration of multiple cameras.** This is highly recommended to provide a wider monitoring coverage that will improve the visibility in different rooms of the house or apartment. This is a big help to detect different hazards more efficiently and gives the residents clearer view of the situation during emergencies.

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# **APPENDIX**

## USER MANUAL

The system ISHA: Integrated Safety and Hazard Alerts System for Home Emergencies offered multiple sensors in one device that included monitoring and early warning detection. By launching the application, users could explore different important sections and guidance for system functionality.

### A. ISHA

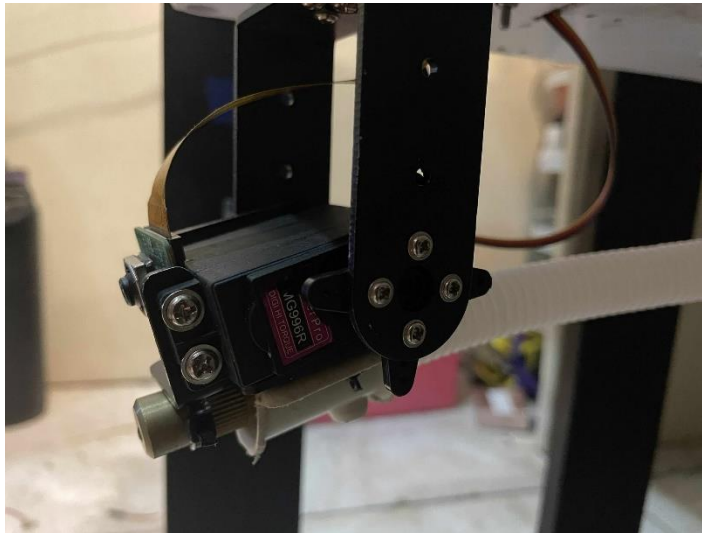


Figure 31. ISHA

- The ISHA system protected its designated space by continuously monitoring for fires, gas leaks, smoke, and earthquakes. It used a network of sensors and cameras and could automatically activate a mist system to suppress a fire.

## B. Launching the application

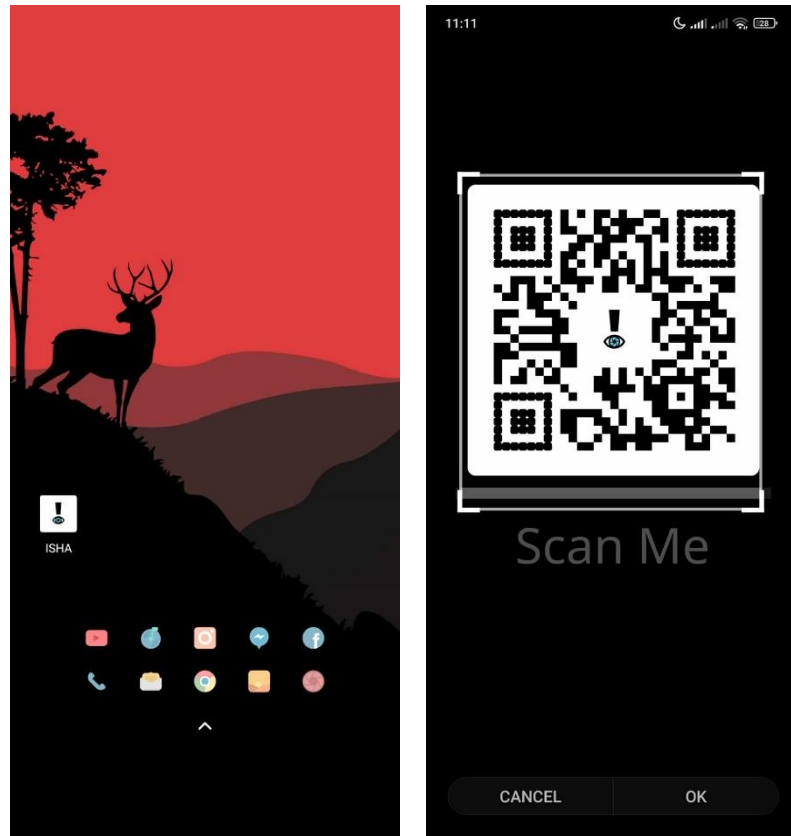


Figure 32. Mobile Application

- The ISHA system was managed through its dedicated mobile application.
- You can download the apps by scanning the QR code with your device's camera.

### C. Logging to Dashboard

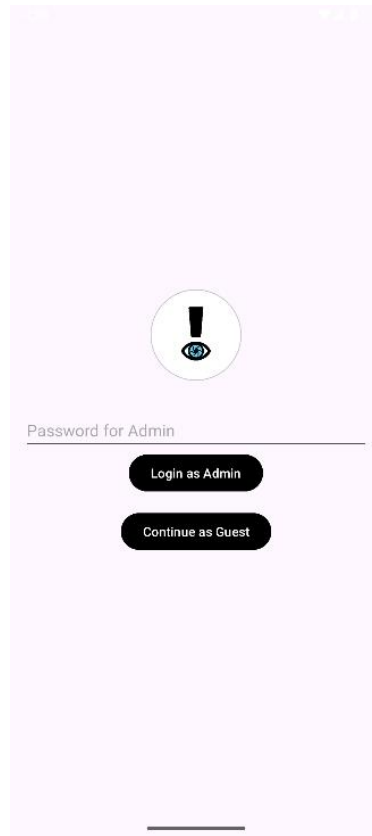


Figure 33. Login to Dashboard

- Clicking the "Continue as Guest" button redirected the user to the login page.
- The user entered a password for the admin passkey.
- Upon successful login, the user was redirected to the admin dashboard.

## D. Navigating Dashboard

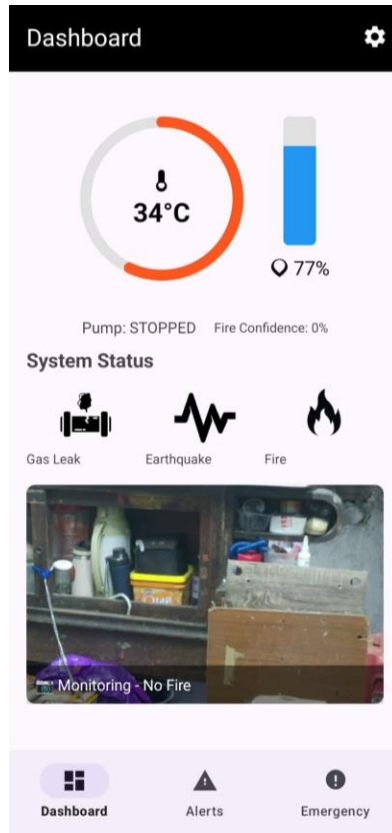


Figure 34. Dashboard

- The user dashboard centrally displayed the current temperature and humidity readings.
- Below this, the user could see the pump status and fire sensitivity.
- The gas leak icon, earthquake icon, and fire icon indicated "Normal" when the situation was safe.
- In the event of an emergency, the system changed the icon with a colored red alert.
- For admin only, a camera feed was available. To view the full-screen live feed, the user clicked the "Open Camera" button.

## E. Admin Camera Control

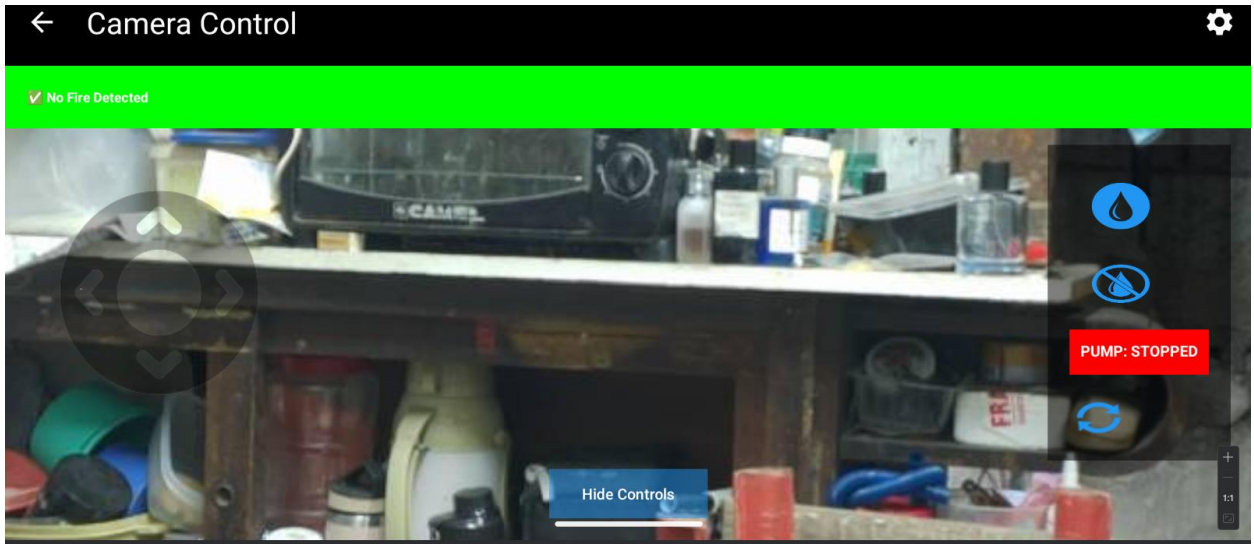


Figure 35. Admin Camera Control Page

- The user could use the camera controls to pan and tilt the view.
- The user could use the Mist Button to turn the misting system on or off.
- The indicator light below the button showed the pump status.
- If a fire was detected, the system automatically outlined it with a red box on the screen.
- The mist mechanism then activated automatically to extinguish the flame.

## G. Viewing Alert Notification



Figure 36. Alert Notification Page

- The system generated notification alerts for three primary hazards.
- Earthquake: Displayed with an orange alert and an earthquake icon.
- Gas Leak: Displayed with a yellow alert and a gas leak icon.
- Fire: Displayed with a red alert and a fire hazard icon.

## H. Calling Emergency Hotline

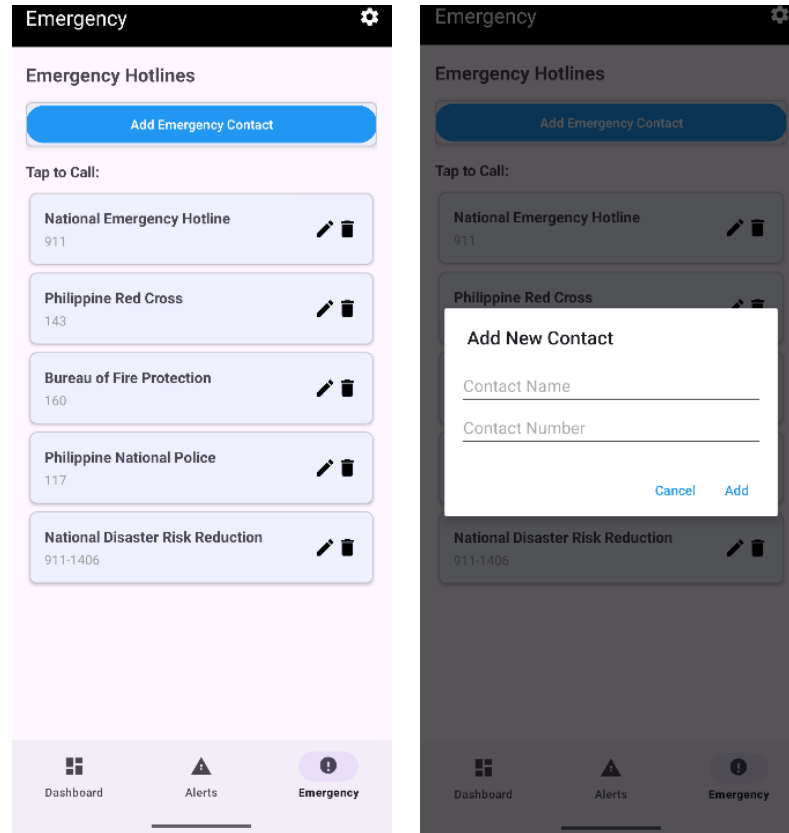


Figure 37. Emergency Hotline Page

- A list of emergency hotlines based on the user's location was provided.
- To call a number, the user simply clicked the call button.
- This action redirected the user to the phone's dialer to initiate the call.
- Users could click "Add Emergency Contact" to add and save their own customized emergency contacts.

## **TECHNICAL MANUAL**

The technical components used in the ISHA system were listed below, with accompanying details on their description, installation, usage instructions, and maintenance.

### **1. Raspberry pi 5 (4Gb)**

The Raspberry Pi 5 (4GB) presented an affordable platform for edge AI applications, capable of running YOLO models for real-time object detection.

#### **Installation/Setup:**

- Install the Raspberry Pi OS onto a microSD card and insert it into the Pi.
- Power the device using a USB-C power adapter and connect any necessary peripherals.
- Complete the initial OS setup.
- Install the required software libraries for the project.
- Integrate external components, such as sensors or actuators as required.

#### **Usage Instructions:**

- Execute the main Python script to initiate the algorithm.
- Executes the algorithm on each data point to fulfill the system's intended function.

#### **Maintenance/Troubleshooting:**

- Maintain adequate device cooling using an active cooler or heatsink.
- Update the system software regularly.
- If the system fails to start, first verify the power supply is connected.

## **2. Raspberry pi Aluminum Armor Case Dual Fan**

The Raspberry Pi Aluminum Armor Case provided physical protection for the device while preventing thermal throttling.

### **Installation/Setup:**

- Carefully place the Raspberry Pi motherboard into the aluminum armor frame.
- Secure the two cooling fans on the case.
- Connect the fans' power cables to the corresponding GPIO pins.

### **Usage Instructions:**

- Upon startup, the dual cooling fans activate automatically.

### **Maintenance/Troubleshooting:**

- Periodically remove any accumulated dust.
- If the fan fails to spin, inspect and secure its power connection.

## **3. MQ-2 Gas Sensor**

The MQ-2 gas sensor was a semiconductor device capable of detecting a hazardous gas, such as smoke, carbon monoxide, and methane.

### **Installation/Setup:**

- Connect the MQ-2 gas sensor to the GIO pins.
- Allow a pre-heating period for the sensor to stabilize.

### **Usage Instructions:**

- Establish a threshold value to differentiate concentrations.
- Read the analog voltage output from the sensor.

#### **Maintenance/Troubleshooting:**

- If readings are inconsistent, verify all wiring connections for loose contacts.
- Recalibrate the threshold values periodically.

#### **4. SW-420 vibration sensor**

The SW-420 was a normally-closed vibration sensor module designed to detect shocks, tremors, and other sudden movements.

#### **Installation/Setup:**

- Connect the sensor to the designated GPIO pin.
- Calibrate the detection sensitivity.

#### **Usage Instructions:**

- A Python script continuously polls the sensor to monitor its state.
- Upon detecting a trigger, the script initiates a predefined emergency protocol.

#### **Maintenance/Troubleshooting:**

- The system's functionality should be validated through periodic sensor tests.
- If frequent false alarms occur, reduce the sensor sensitivity.
- If the sensor fails to trigger, inspect and verify all wiring connections.

#### **5. DHT22 Temperature/Humidity Sensor**

The DHT22 was a low-cost digital sensor that measured ambient temperature and humidity with high accuracy and long-term stability.

#### **Installation/Setup:**

- Connect the DHT22 sensor to the appropriate GPIO pins.
- Enable the I2C interface via configuration settings.

#### **Usage Instructions:**

- Import the necessary library.
- Implement conditional logic to analyze the incoming sensor data.

#### **Maintenance/Troubleshooting:**

- If inconsistent readings, place the sensor away from heat sources.
- If sensor not detected, Verify the correctness of all wiring connections.
- Periodically inspect the sensor for any accumulated dust or debris.

### **6. Relay Module**

A relay module functioned as an electrically operated switch, enabling a low-power signal from a microcontroller to control a high-power device.

#### **Installation/Setup:**

- Connect to the designated GPIO pins.
- Attach the high-power device to the output terminals.

#### **Usage Instructions:**

- Installed necessary Python library within the programming environment.
- A high signal from the GPIO pin energizes the relay's electromagnet.

#### **Maintenance/Troubleshooting:**

- If device stays on, verify that the project code correctly.
- If relay does not activate, confirm that all wiring connections are correctly seated.

## **7. Raspberry Pi Camera Module 3**

The Raspberry Pi Camera Module 3 was a compact, high-resolution peripheral engineered for Raspberry Pi single-board computers.

### **Installation/Setup:**

- Carefully insert the camera's flexible ribbon cable into the CSI port.
- Type the command `sudo raspi-config` to launch the configuration tool.
- Navigate to Interface Options > Camera and enable it.

### **Usage Instructions:**

- The camera module is initialized by the main Python script.
- The initialized camera captures a real-time video stream.

### **Maintenance/Troubleshooting:**

- If blurry footage, clean it gently with a microfiber cloth.
- If camera not detected, verify that the ribbon cable is fully inserted.
- If system crashes or freezes, consider using a lower resolution or a lighter-weight model.

## **8. Active Alarm Buzzer Module**

The Active Alarm Buzzer Module functions as an audio output device, generating a loud, audible alert.

### **Installation/Setup:**

- Connect to the designated GPIO pins.
- Install the required library in your Python environment.

#### **Usage Instructions:**

- Control the buzzer programmatically.
- The main Python script continuously monitors the system state.

#### **Maintenance/Troubleshooting:**

- Periodically test the buzzer functionality.
- If the buzzer produces no sound, verify that all wiring connections are secure.
- If the sound is weak or distorted, ensure the buzzer is receiving adequate power.

### **9. MG996R Servos**

The MG996R was a high-torque digital servo motor which made it well-suited for applications demanding substantial force.

#### **Installation/Setup:**

- The servo motor is connected to the PWM pin.
- Use a sufficiently powerful external power supply.

#### **Usage Instructions:**

- Install the necessary software library.
- A Python script manages the servo's control logic.

#### **Maintenance/Troubleshooting:**

- Avoid forcing the servo horn beyond its mechanical limits.
- If the servo becomes unresponsive, verify the stability of the external power.

## **10. Micro SD Card (32GB)**

A 32GB Micro SD card functions as the primary boot device and storage medium.

### **Installation/Setup:**

- Flash the operating system to the microSD card.
- Insert the card into the microSD card slot.

### **Usage Instructions:**

- The system continuously uses the card for runtime operations.

### **Maintenance/Troubleshooting:**

- Periodically a backup once the system is fully configured and stable.
- If the system fails to boot, consider SD card corruption.

## **11. 12V 10A Power Supply Unit (PSU)**

The 12V 10A Power Supply Unit (PSU) is a switched-mode device that converts 110V/220V AC household current to a stable 12V DC output.

### **Installation/Setup:**

- Mount the Power Supply Unit (PSU) in a secure, well-ventilated location.
- Connect all terminals, ensuring correct polarity.

### **Usage Instructions:**

- The system operates in a continuous to provide uninterrupted emergency monitoring.

### **Maintenance/Troubleshooting:**

- Perform regular inspections of the PSU.
- Ensure that all ventilation slots remain unobstructed.

- If the system loses power, first verify that the AC input is active.

## **12. Water Pressure Nozzle**

The Water Pressure Nozzle was a specialized firefighting attachment engineered for integration with a standard water hose and a solenoid valve.

### **Installation/Setup:**

- Securely fasten the Water Pressure Nozzle.

### **Usage Instructions:**

- It is designed for fully autonomous operation.

### **Maintenance/Troubleshooting:**

- Inspect all fluid connection points.
- Perform routine functional tests.

## **13. Diaphragm Water Pump**

A diaphragm water pump was a positive displacement pump to move fluid via an electrically operated mechanism.

### **Installation/Setup:**

- Install the pump in an accessible location near a water source.
- Connect to power the unit using a 12V DC power supply.

### **Usage Instructions:**

- The pump operates under the command of a central control system.

#### **Maintenance/Troubleshooting:**

- Periodically inspect the diaphragm for signs of wear.
- Avoid dry operation to prevent diaphragm damage.

#### **14. LM2596S Voltage Regulators**

The LM2596S was a monolithic integrated circuit that provides a step-down voltage conversion function.

#### **Installation/Setup:**

- Connect the positive terminal of the power source.
- Use a small screwdriver to carefully adjust the potentiometer.

#### **Usage Instructions:**

- Ensure the input voltage remains within the module's specified maximum rating.

#### **Maintenance/Troubleshooting:**

- Periodically verify the input and output voltages.
- If the regulator becomes excessively hot, immediately inspect the circuit.

#### **15. PCA9685 Servo Controller**

The PCA9685 was an I2C-bus controlled servo driver that could independently operate up to 16 servo motors or PWM devices.

#### **Installation/Setup:**

- Connect to the GPIO pins.
- Install the necessary Python libraries.

#### **Usage Instructions:**

- Translate the commands from the algorithm.

#### **Maintenance/Troubleshooting:**

- Periodically inspect all physical connections.
- If the servo module is unresponsive, verify that the I2C protocol is enabled.

### **16. 12V Fans**

The 12V DC fan was designed to generate airflow for ventilation and active cooling purposes.

#### **Installation/Setup:**

- Connect the 12V fan to a compatible 12V power supply.

#### **Usage Instructions:**

- Fan operation is controlled automatically by commands.

#### **Maintenance/Troubleshooting:**

- Perform periodic maintenance by cleaning the fan blades.
- If the fan fails to activate, first verify the functionality of the power source.

Attached is the schematic made for the hardware of the ISHA system.

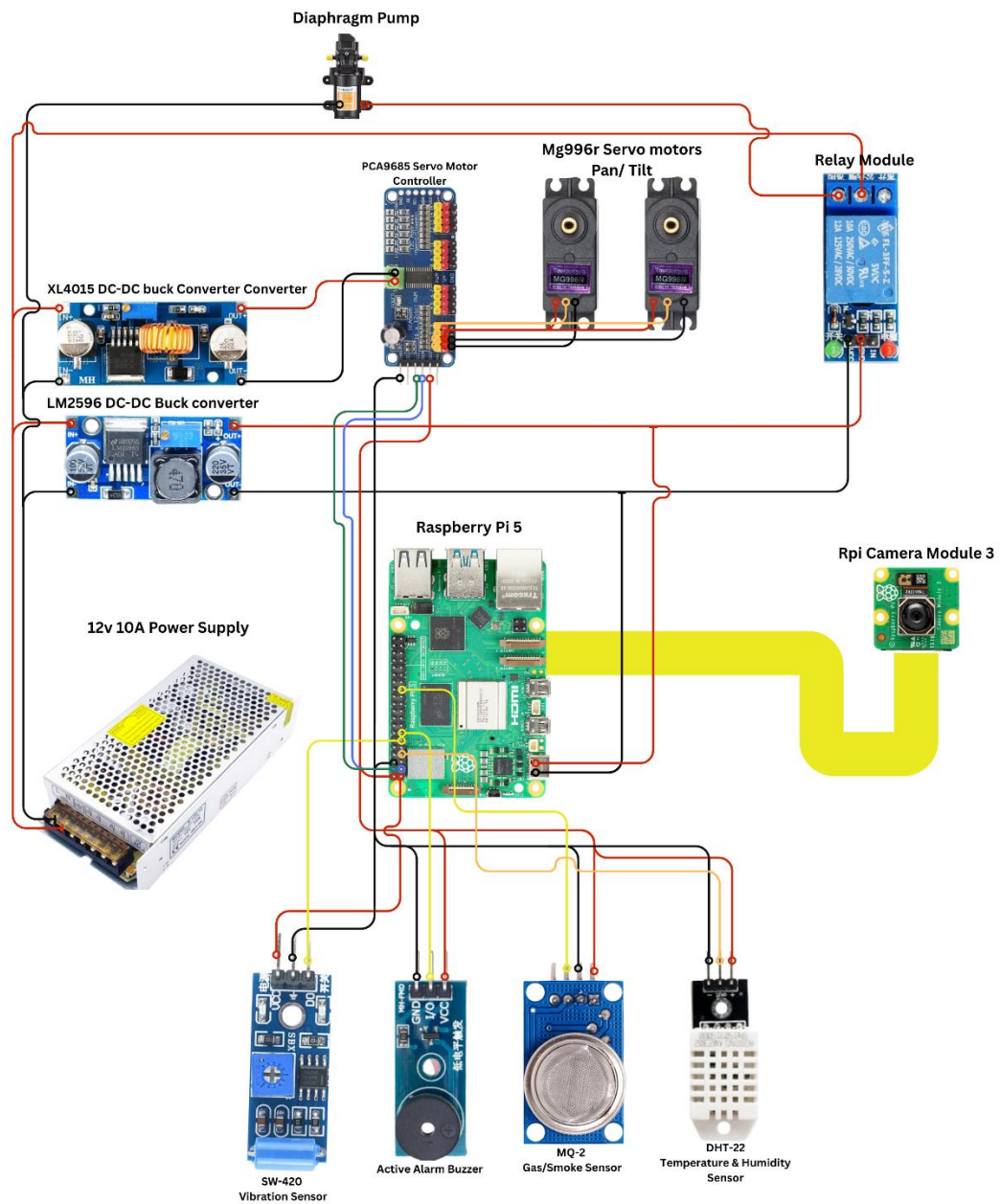


Figure 38. Schematic

## USER INTERFACE

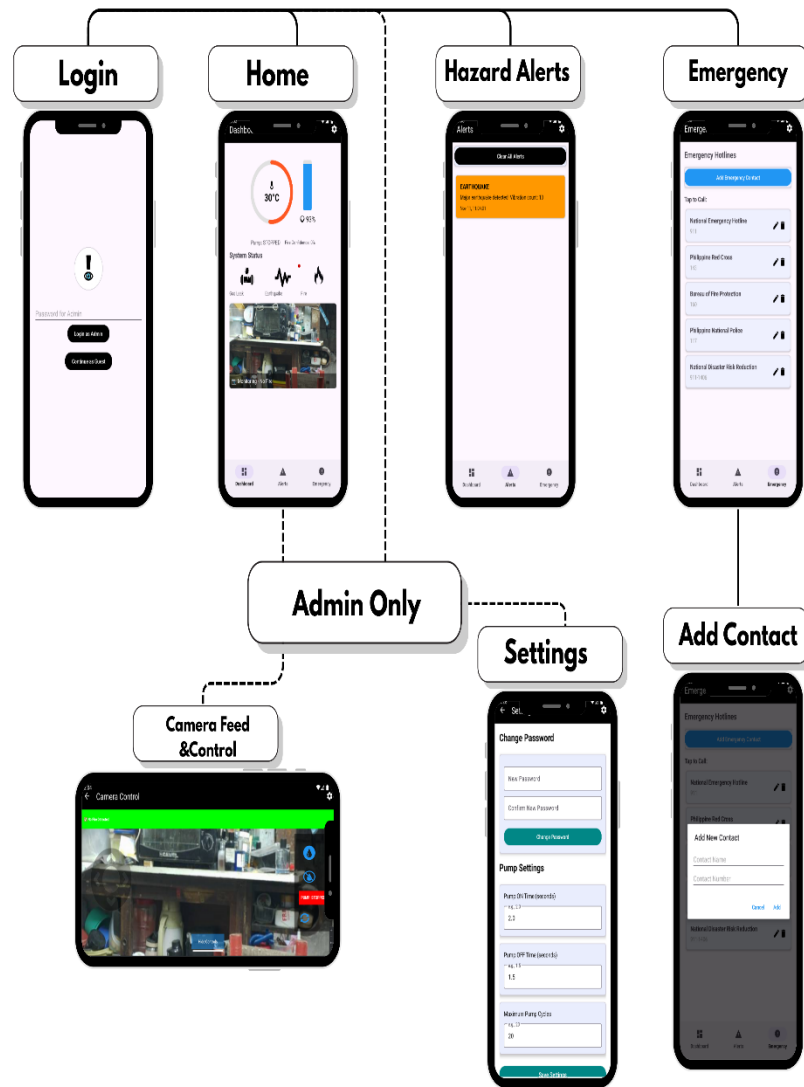


Figure 39. User Interface

## ENVIRONMENTAL OBSERVATION

The researchers carried out an environmental assessment in the neighborhood where they intended to introduce the ISHA system. They observed the existing safety conditions inside the homes and apartments which mainly focused on how the residents currently detected different hazards such as fire, smoke, gas leaks, and earthquakes. Researchers also observed the existing communications that were available in the community. This included the dependability of the internet and mobile network signals in the area, which was essential for timely alerts. The team gave special attention to how residents reacted to emergencies and the difficulties they might face when an incident occurred. Following the observation phase, the researchers developed a survey questionnaire designed for residents and tenants in the area to collect information regarding their safety requirements, awareness of emergencies, and willingness to embrace an IoT-based alert system.

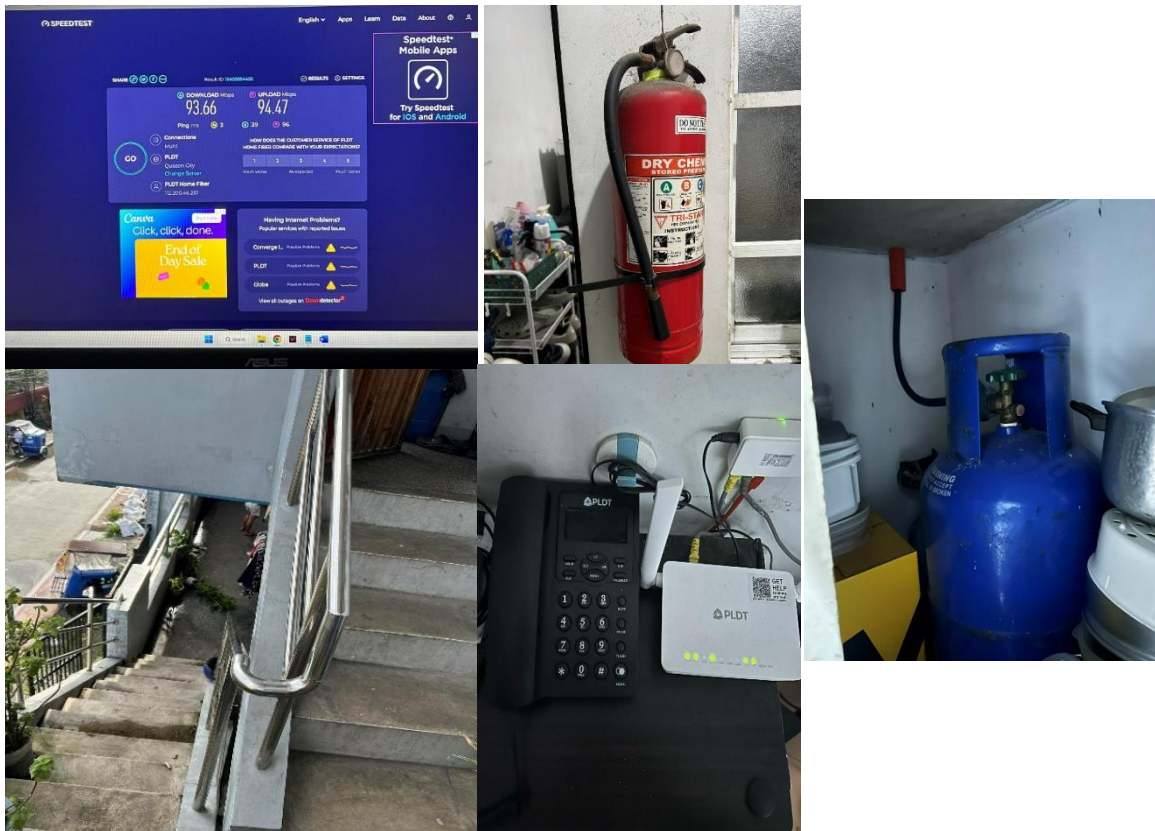


Figure 40. Environmental Observation

## SURVEY QUESTIONNAIRE DISTRIBUTION

Survey questionnaires were developed to collect residents' views on their current readiness for emergencies, and the efficiency of the hazard detection measures in their homes. This effort aimed to assess the need for an IoT-based multi-hazard alert system and to gain insight into the public's awareness and willingness to adopt technologies such as sensors, mobile notifications, and real-time monitoring.

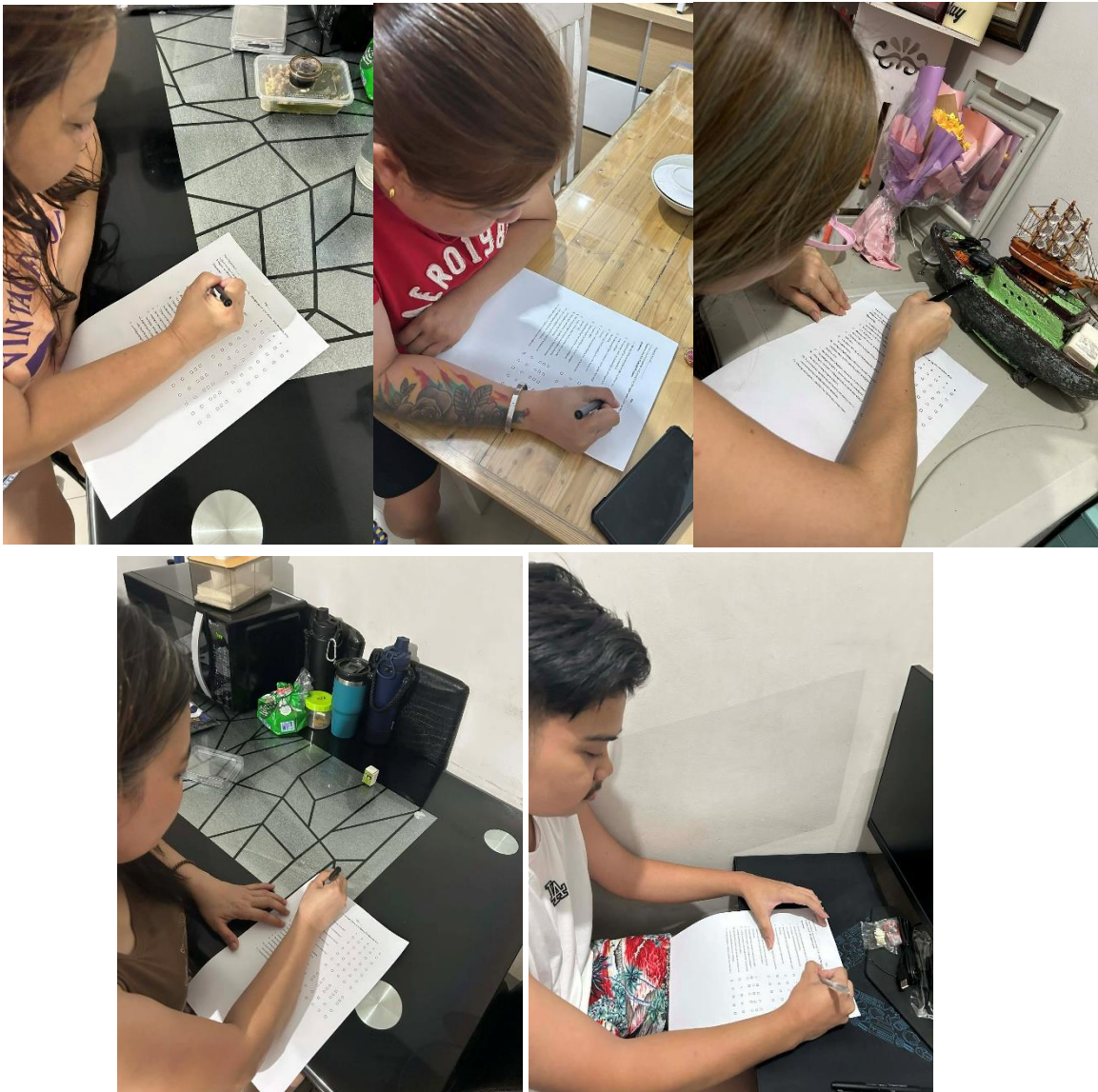


Figure 41. Survey Questionnaire Distribution

## PROTOTYPE DEPLOYMENT AND USER FEEDBACK GATHERING

A different set of survey questions was developed to collect responses from participants about the suggested ISHA alert system for home emergencies. The findings offered a valuable understanding of their approval of the device's characteristics, which were elaborated on in the Results and Discussion section.



Figure 42. Prototype Deployment and User Feedback Gathering

Name (optional): \_\_\_\_\_

Age\_\_\_\_\_

I. Check the boxes appropriately using **1** for **Strongly Agree**, **2** for **Agree**, **3** for **Neutral**, **4** for **Disagree**, and **5** for **Strongly Disagree**.

| Questions:                                                                                                  | 1                        | 2                        | 3                        | 4                        | 5                        |
|-------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| 1. Having safety devices like smoke alarms or gas detectors at home is essential.                           | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 2. Immediate notification during a fire emergency is very important.                                        | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 3. Knowing what to do during an earthquake is crucial for safety.                                           | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 4. Detecting potential gas leaks in the home is a concern.                                                  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 5. A device that can automatically put out a small fire is a good idea.                                     | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 6. Using a single device that can detect multiple hazards such as fire, earthquake, and gas is trustworthy. | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 7. Using a smart device can make a home safer.                                                              | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 8. Receiving emergency alerts on phone is helpful.                                                          | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 9. Having a camera to visually check the home status during an emergency is useful.                         | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 10. A good safety system must be easy for everyone to use, including the elderly and children.              | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 11. Having emergency hotline numbers are important.                                                         | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

## ENVIRONMENTAL OBSERVATION RESULTS

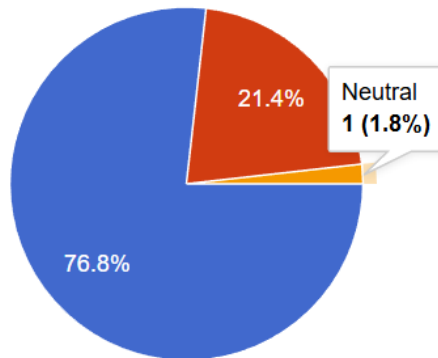
The color basis of the survey is represented by:



*Figure 43: Color basis*

### Pre-survey

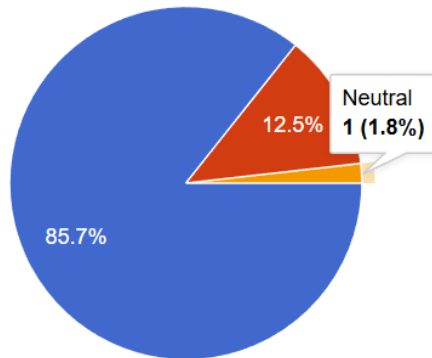
1. Having safety devices like smoke alarms or gas detectors at home is essential.



*Figure 44: Survey Responses No. 1*

With a total of 56 responses, 76.8% of responders strongly agreed, 21.4% agreed, and 1.8% were neutral that having safety devices like smoke alarms or gas detectors at home was essential.

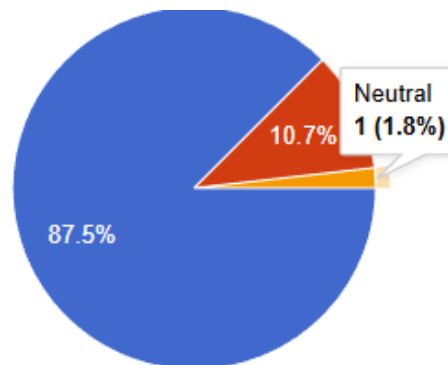
2. Immediate notification during a fire emergency is very important.



**Figure 45: Survey Responses No. 2**

Out of 56 respondents, 85.7% strongly agree, 12.5% agree, and 1.8% choose neutral on the importance of immediate notification during fire emergencies.

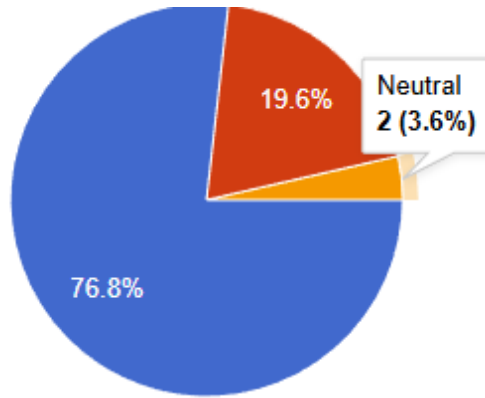
3. Knowing what to do during an earthquake is crucial for safety.



**Figure 46: Survey Responses No. 3**

Out of 56 respondents, 85.7% strongly agreed, 12.5% agreed, and 1.8% chose neutral on the importance of immediate notification during fire emergencies. 3. Knowing what to do during an earthquake was crucial for safety.

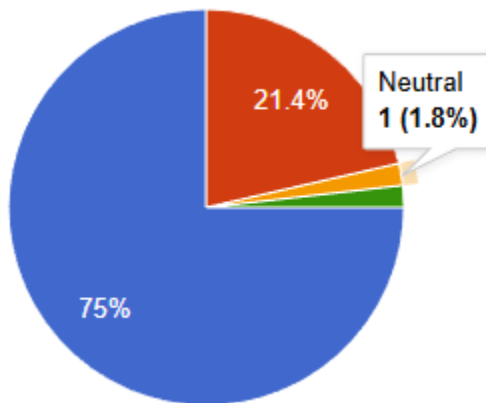
4. Detecting potential gas leaks in the home is a concern.



**Figure 47: Survey Responses No. 4**

Out of 56 respondents, 76.8% strongly agreed, 19.6% agreed, and 3.6% were neutral on detecting potential gas leaks in their home are their concern.

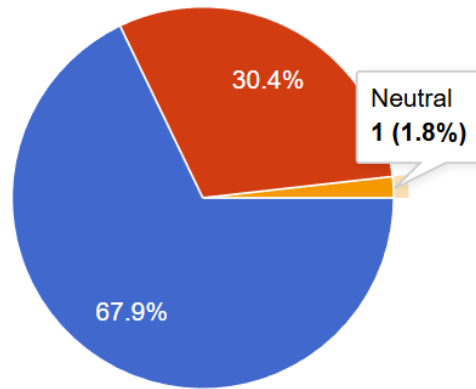
5. A device that can automatically put out a small fire is a good idea.



**Figure 48: Survey Responses No. 5**

Out of 56 respondents, 75% strongly agreed, 21.4% agreed, 1.8% were neutral, and 1.8% disagreed that the device that could automatically put out a small fire was a good idea.

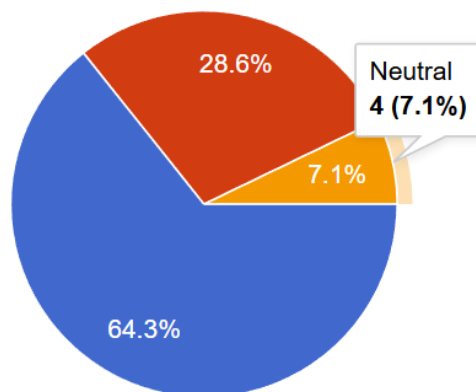
6. Using a single device that can detect multiple hazards such as fire, earthquake, and gas is trustworthy.



**Figure 49: Survey Responses No. 6**

Out of 56 respondents, 67.9% strongly agreed, 30.4% agreed, and 1.8% were neutral about using a single device that could detect various hazards such as fire, earthquake, and gas.

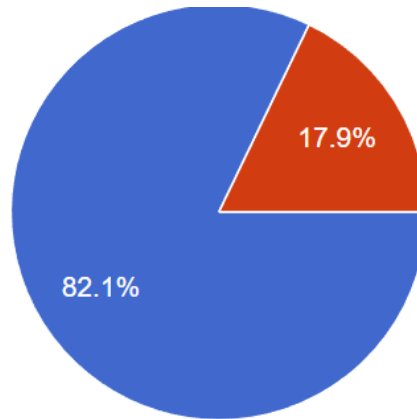
7. Using a smart device can make a home safer.



**Figure 50: Survey Responses No. 7**

Out of 56 respondents, 64.3% strongly agreed, 28.6% agreed, and 7.1% were neutral about when using a smart device could make their home safer.

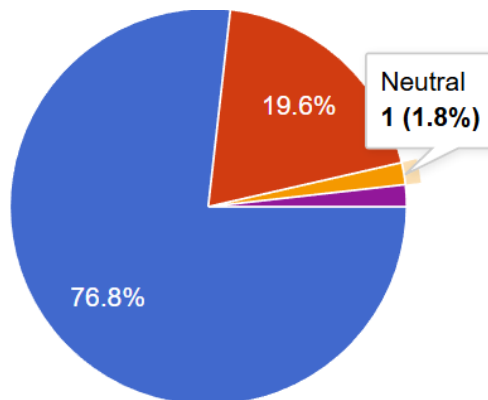
8. Receiving emergency alerts on phone is helpful.



**Figure 51: Survey Responses No. 8**

Out of 56 respondents, 82.1% strongly agreed and 17.9% agreed that receiving emergency alerts on their phone was helpful.

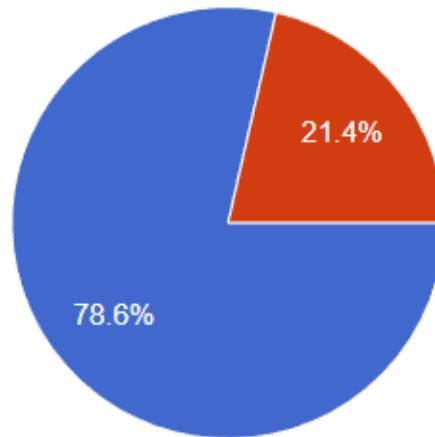
9. Having a camera to visually check the home status during an emergency is useful.



**Figure 52: Survey Responses No. 9**

Out of 56 respondents, 76.8% strongly agreed, 19.6% agreed, 1.8% were neutral, and 1.8% strongly disagreed that having a camera to visually check the home status during an emergency was useful.

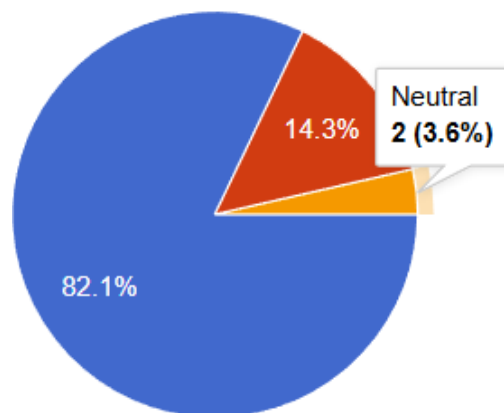
10. A good safety system must be easy for everyone to use, including the elderly and children.



**Figure 53: Survey Responses No. 10**

Out of 56 respondents, 78.6% strongly agreed and 21.4% agreed that having a good system had to be easy for everyone to use, including the elderly and children.

11. Having emergency hotline numbers are important.

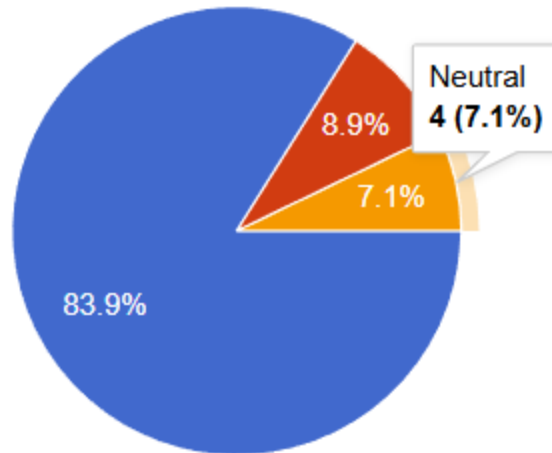


**Figure 54: Survey Responses No. 11**

Out of 56 respondents, 82.1% strongly agreed, 14.3% agreed, and 3.6% were neutral about having emergency hotline numbers were important.

## Post-survey

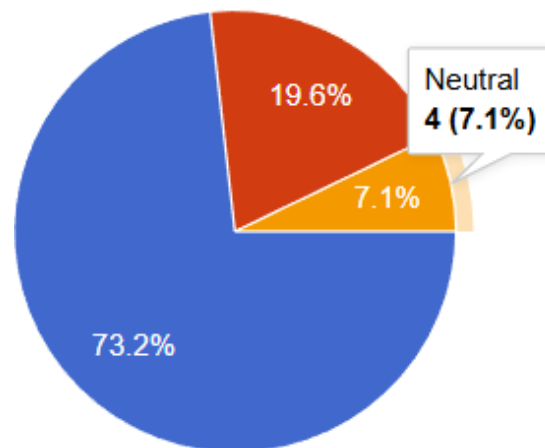
1. The system's ability to detect different hazards like fire, gas, and earthquakes makes me feel safer.



**Figure 55: Survey Responses No. 1**

Out of 56 respondents, 83.9% strongly agreed, 8.9% agreed, and 7.1% were neutral about the system's ability to detect different hazards that made them feel safer.

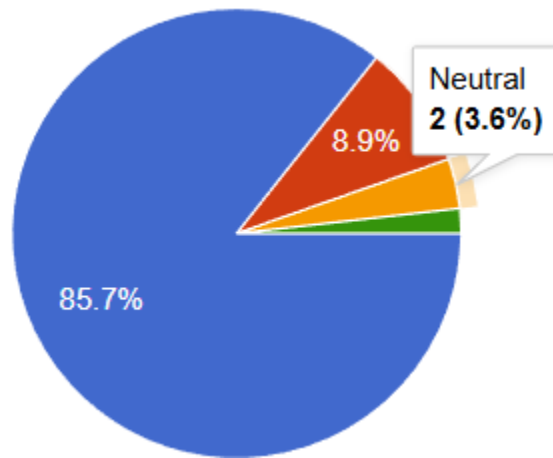
2. The idea of an automatic water spray to put out a small fire is reassuring.



**Figure 56: Survey Responses No. 2**

Out of 56 respondents, 73.2% strongly agreed, 19.6% agreed, and 7.1% were neutral about the idea of an automatic water spray to put out small fires. 3. Getting an early warning for an earthquake would have been very helpful.

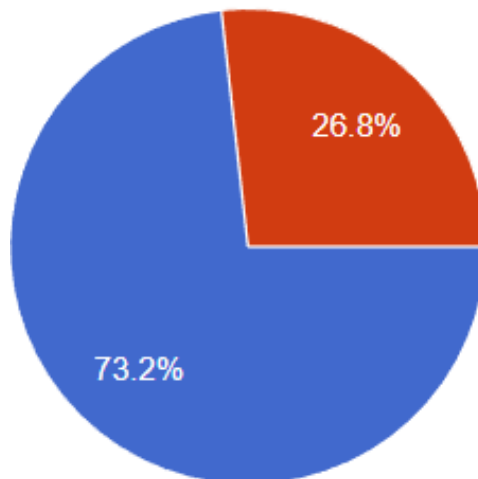
3. Getting an early warning for an earthquake would be very helpful.



**Figure 57: Survey Responses No. 3**

Out of 56 respondents, 85.7% strongly agreed, 8.9% agreed, 3.6% were neutral, and 1.8% disagreed that getting an early warning for an earthquake was helpful.

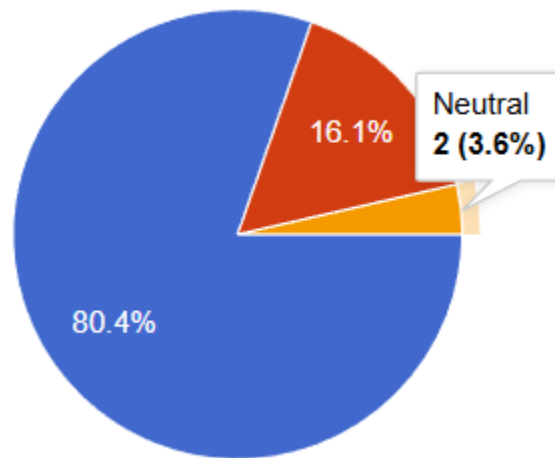
4. Having one device for multiple hazards is better than having many separate devices.



**Figure 58: Survey Responses No. 4**

Out of 56 respondents, 73.2% strongly agreed and 26.8% agreed that having one device with detection of multiple hazards was better than having many separate devices. 5. The system's camera feature for checking the home was a valuable addition.

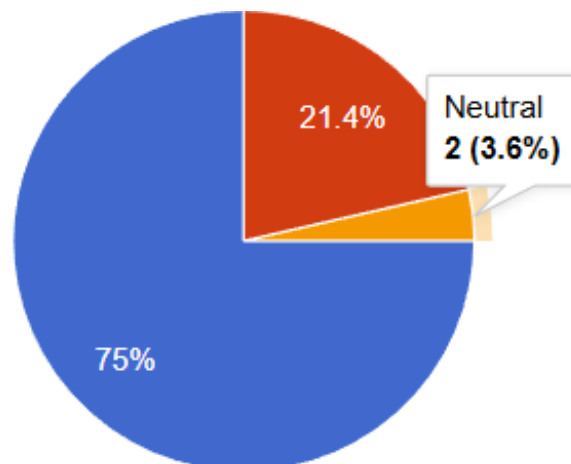
5. The system's camera feature for checking the home is a valuable addition.



**Figure 59: Survey Responses No. 5**

Out of 56 respondents, 80.4% strongly agreed, 16.1% agreed, and 3.6% were neutral about having a camera feature for checking and monitoring the home as a valuable addition. 6. This system could help reduce damage to property and prevent injuries during emergencies.

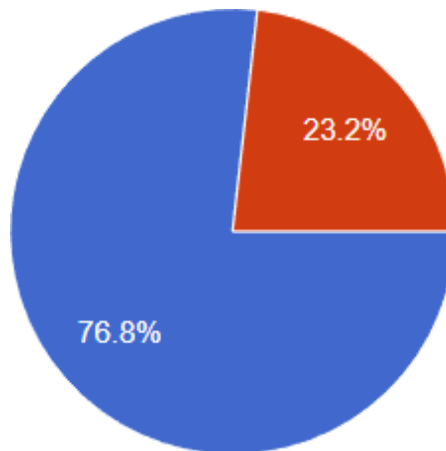
6. This system could help reduce damage to property and prevent injuries during emergencies.



**Figure 60: Survey Response No. 6**

Out of 56 respondents, 75% strongly agreed, 21.4% agreed, and 3.6% were neutral about whether the system would be helpful for reducing damage to the property and avoiding injuries during emergencies.

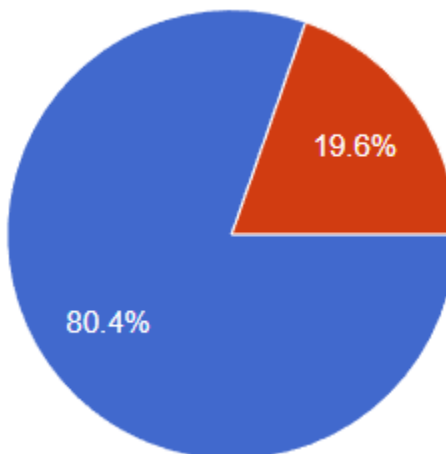
7. I feel more informed about home safety after learning about this system.



**Figure 61: Survey Responses No. 7**

Out of 56 respondents, 76.8% strongly agreed and 23.2% agreed that they gained more knowledge about home safety after their experience on the system. 8. Receiving alerts directly to my phone during an emergency was a useful feature.

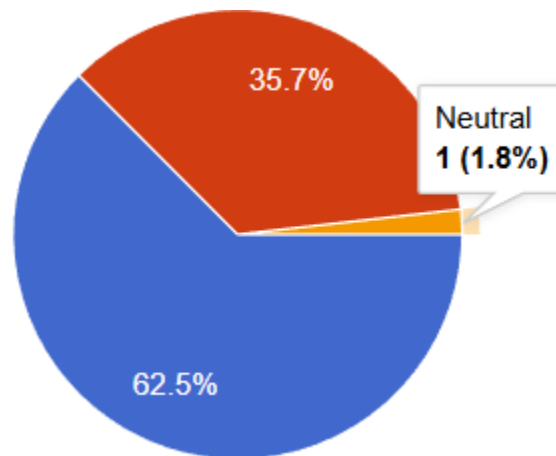
8. Receiving alerts directly to my phone during an emergency is a useful feature.



**Figure 62: Survey Responses No. 8**

Out of 56 respondents, 80% strongly agreed and 19.6% agreed that receiving alerts directly to their phone during an emergency was a useful feature.

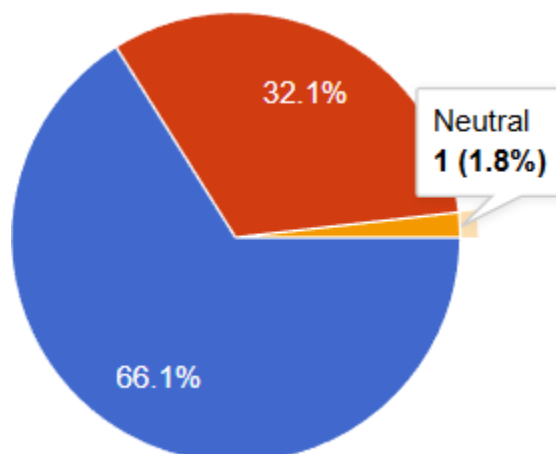
9. The mobile app for monitoring the home seems user-friendly.



**Figure 63: Survey Responses No. 9**

Out of 56 respondents, 62.5% strongly agreed, 35.7% agreed, and 1.8% were neutral about the monitoring function of the mobile app being user-friendly. 10. The way the system sent app alerts was reliable.

10. The way the system sends app alerts is reliable.



**Figure 64: Survey Response No. 10**

Out of 56 respondents, 66.1% strongly agreed, 32.1% agreed, and 1.8% were neutral about the way the system sent alerts on the application was reliable.

The following questions were asked to the respondents for them to respond with a freeform answer.

*Do you have any feedback or recommendation about our system? If yes, please leave suggestions.*

The feedback collected from users regarding the ISHA prototype was predominantly positive, confirming the system's potential to effectively address important home safety needs. Of the 56 respondents interviewed, 39 individuals (70%) expressed strong satisfaction with the ISHA device and its mobile application. These users particularly appreciated the system's core functionalities, its intuitive user interface, integrated hazard detection capabilities, clear alert mechanisms, and the convenience of remote monitoring. The remaining 17 respondents (30%) also provided valuable input, which included minor suggestions for further enhancements to the system. Their primary recommendations involved the integration of notification alerts for increased notification redundancy and minor refinements to the mobile application's user interface for improved accessibility. These suggestions indicated a desire for continuous enhancement and robust communication options, rather than pointing to fundamental issues with the system's design. This feedback provided clear directions for future updates, focusing on optimizing the user experience and bolstering communication resilience.

## USER FEEDBACK

The researchers surveyed to further strengthen and gather data to determine the effectiveness of the study. The target respondents of the researchers were apartments with members of a family who were staying together, since they had extensive experience. Thus, the tables presented were the summaries of the combined data gathered through the survey.

### Profile of the Respondents

| Age Group    | Number Of Respondents | Percentage (%) |
|--------------|-----------------------|----------------|
| 41 and above | 4                     | 7.1%           |
| 31 to 40     | 1                     | 1.8%           |
| 21 to 30     | 42                    | 75%            |
| 20 and below | 10                    | 17.9%          |
| <b>TOTAL</b> | <b>56</b>             | <b>100%</b>    |

*Table 4. Age Profile of the Respondents*

Table 4 presented the distribution of respondents according to their age brackets. Out of the total 56 respondents, 4 individuals (7.1%) belonged to the 41 years and above age group. Only 1 respondent (1.8%) fell within the 31–40 age bracket. The majority of respondents, comprising 42 individuals (75.0%), were within the 21–30 age range, indicating that this age group had the highest participation in the survey. Meanwhile, 10 respondents (17.9%) were aged 20 years and below.

| Emerging Technology Experience                                       | Yes       | (%)         | No       | (%)       |
|----------------------------------------------------------------------|-----------|-------------|----------|-----------|
| I am familiar with using technology devices like cellphones          | 56        | 100%        | 0        | 0%        |
| I am aware of Smart technology as a means of help in our daily lives | 56        | 100%        | 0        | 0%        |
| <b>TOTAL</b>                                                         | <b>56</b> | <b>100%</b> | <b>0</b> | <b>0%</b> |

*Table 5 Technology Experience Profile of the Respondents*

Table 5 indicated that all respondents (100%) were familiar with the use of technology, particularly smart devices. Moreover, they demonstrated a high level of awareness regarding emerging technologies and their potential to assist in various aspects of their daily lives.

### **Feedback from the Respondents**

The Likert Scale was used to measure the respondents' experience upon using the proposed system, ISHA: Integrated Safety and Hazard Alert System for Home Emergencies. The scale ranges from the following: (5) Strongly Agree, (4) Agree, (3) Neutral, (2) Disagree, and (1) Strongly Disagree.

| <b>Weight</b> | <b>Limits</b>      | <b>Descriptive Rating</b> |
|---------------|--------------------|---------------------------|
| <b>5</b>      | <b>4.50 – 5.00</b> | <b>Strongly Agree</b>     |
| <b>4</b>      | <b>3.50 – 4.49</b> | <b>Agree</b>              |
| <b>3</b>      | <b>2.50 – 3.49</b> | <b>Neutral</b>            |
| <b>2</b>      | <b>1.50 – 2.49</b> | <b>Disagree</b>           |
| <b>1</b>      | <b>1.00 – 1.49</b> | <b>Strongly Disagree</b>  |

*Table 6 Likert Scale Range of Scores*

Table 6 showed the ranges of scores used to describe the feedback of the respondents upon using ISHA: Integrated Safety and Hazard Alert System for Home Emergencies. The basis of scores was obtained from the result of pre-survey and post-survey questionnaires.

**Question #1: The system's ability to detect different hazards like fire, gas, and earthquakes makes me feel safer.**

| <b>Range</b>                      | <b>No. of Responses</b> | <b>Percentage (%)</b> |
|-----------------------------------|-------------------------|-----------------------|
| Strongly Agree                    | 47                      | 83.9%                 |
| Agree                             | 5                       | 8.9%                  |
| Neutral                           | 4                       | 7.1%                  |
| Disagree                          | 0                       | 0%                    |
| Strongly Disagree                 | 0                       | 0%                    |
| <b>TOTAL</b>                      | 56                      | 100%                  |
| <b>MEAN:4.77 – Strongly Agree</b> |                         |                       |

*Table 7 Question No.1 Responses*

Table 7 presented the responses of 56 participants. Among them, 47 respondents, equivalent to 83.9%, strongly agreed with the statement, while 5 respondents or 8.9% agreed, and 4 respondents or 7.1% remained neutral. The responses had a mean score of 4.77, which fell within the range of strongly agreed. This result indicated that the respondents believed that individuals felt much safer when the systems had the capability to detect potential hazards such as fire, gas, and earthquake.

**Question #2: The idea of an automatic water spray to put out a small fire is reassuring.**

| <b>Range</b>                      | <b>No. of Responses</b> | <b>Percentage (%)</b> |
|-----------------------------------|-------------------------|-----------------------|
| Strongly Agree                    | 41                      | 73.2%                 |
| Agree                             | 11                      | 19.6%                 |
| Neutral                           | 4                       | 7.1%                  |
| Disagree                          | 0                       | 0%                    |
| Strongly Disagree                 | 0                       | 0%                    |
| <b>TOTAL</b>                      | 56                      | 100%                  |
| <b>MEAN:4.66 – Strongly Agree</b> |                         |                       |

*Table 8 Question No.2 Responses*

Table 8 presented the responses of 56 participants. Among them, 41 respondents, equivalent to 73.2%, strongly agreed with the statement, while 11 respondents or 19.6% agreed, and 4 respondents or 7.1% remained neutral. The responses had a mean score of 4.66, which fell within the range of strongly agreed. This result indicated that the respondents were reassured having an automatic water spray that put out a small fire.

**Question #3: Getting an early warning for an earthquake would be very helpful.**

| <b>Range</b>                      | <b>No. of Responses</b> | <b>Percentage (%)</b> |
|-----------------------------------|-------------------------|-----------------------|
| Strongly Agree                    | 48                      | 85.7%                 |
| Agree                             | 5                       | 8.9%                  |
| Neutral                           | 2                       | 3.6%                  |
| Disagree                          | 1                       | 1.8%                  |
| Strongly Disagree                 | 0                       | 0%                    |
| <b>TOTAL</b>                      | 56                      | 100%                  |
| <b>MEAN:4.79 – Strongly Agree</b> |                         |                       |

*Table 9 Question No.3 Responses*

Table 9 presented the responses of 56 participants. Among them, 48 respondents, equivalent to 85.7%, strongly agreed with the statement, while 5 respondents or 8.9% agreed, 2 respondents or 3.6% remained neutral and 1 respondent disagreed or 1.8%. The responses had a mean score of 4.79, which fell within the range of strongly agreed. This result indicated that having early warning for an earthquake was helpful.

**Question #4: Having one device for multiple hazards is better than having many separate devices.**

| <b>Range</b>                      | <b>No. of Responses</b> | <b>Percentage (%)</b> |
|-----------------------------------|-------------------------|-----------------------|
| Strongly Agree                    | 41                      | 73.2%                 |
| Agree                             | 15                      | 26.8%                 |
| Neutral                           | 0                       | 0%                    |
| Disagree                          | 0                       | 0%                    |
| Strongly Disagree                 | 0                       | 0%                    |
| <b>TOTAL</b>                      | 56                      | 100%                  |
| <b>MEAN:4.73 – Strongly Agree</b> |                         |                       |

*Table 10 Question No.4 Responses*

Table 10 presented the responses of 56 participants. Among them, 41 respondents, equivalent to 73.2%, strongly agreed with the statement and 15 respondents or 26.8% agreed. The responses had a mean score of 4.73, which fell within the range of strongly agreed. This result indicated that having one device for multiple sensors was better than having many separate devices.

**Question #5: The system's camera feature for checking the home is a valuable addition.**

| <b>Range</b>                      | <b>No. of Responses</b> | <b>Percentage (%)</b> |
|-----------------------------------|-------------------------|-----------------------|
| Strongly Agree                    | 45                      | 80.4%                 |
| Agree                             | 9                       | 16.1%                 |
| Neutral                           | 2                       | 3.6%                  |
| Disagree                          | 0                       | 0%                    |
| Strongly Disagree                 | 0                       | 0%                    |
| <b>TOTAL</b>                      | 56                      | 100%                  |
| <b>MEAN:4.77 – Strongly Agree</b> |                         |                       |

*Table 11 Question No.5 Responses*

Table 11 presented the responses of 56 participants. Among them, 45 respondents, equivalent to 80.4%, strongly agreed with the statement, while 9 respondents or 16.1% agreed, and 2 respondents or 3.6% were neutral. The responses had a mean score of 4.77, which fell within the range of strongly agreed. This result indicated that having a camera for monitoring the home was a valuable addition.

**Question #6: This system could help reduce damage to property and prevent injuries during emergencies.**

| <b>Range</b>                      | <b>No. of Responses</b> | <b>Percentage (%)</b> |
|-----------------------------------|-------------------------|-----------------------|
| Strongly Agree                    | 42                      | 75%                   |
| Agree                             | 12                      | 21.4%                 |
| Neutral                           | 2                       | 3.6%                  |
| Disagree                          | 0                       | 0%                    |
| Strongly Disagree                 | 0                       | 0%                    |
| <b>TOTAL</b>                      | 56                      | 100%                  |
| <b>MEAN:4.71 – Strongly Agree</b> |                         |                       |

*Table 12 Question No.6 Responses*

Table 12 presented the responses of 56 participants. Among them, 42 respondents, equivalent to 75%, strongly agreed with the statement, while 12 respondents or 21.4% agreed, and 2 respondents or 3.6% were neutral. The responses had a mean score of 4.71, which fell within the range of strongly agreed. This result indicated that the system was helpful for reducing damage to property and preventing injuries during emergencies.

**Question #7: I feel more informed about home safety after learning about this system.**

| <b>Range</b>                      | <b>No. of Responses</b> | <b>Percentage (%)</b> |
|-----------------------------------|-------------------------|-----------------------|
| Strongly Agree                    | 43                      | 76.8%                 |
| Agree                             | 13                      | 23.2%                 |
| Neutral                           | 0                       | 0%                    |
| Disagree                          | 0                       | 0%                    |
| Strongly Disagree                 | 0                       | 0%                    |
| <b>TOTAL</b>                      | 56                      | 100%                  |
| <b>MEAN:4.77 – Strongly Agree</b> |                         |                       |

*Table 13 Question No.7 Responses*

Table 13 presented the responses of 56 participants. Among them, 43 respondents, equivalent to 76.8%, strongly agreed with the statement and 13 respondents or 23.2% agreed. The responses had a mean score of 4.77, which fell within the range of strongly agreed. This result indicated that they gained knowledge about home safety after their experience on the system.

**Question #8: Receiving alerts directly to my phone during an emergency is a useful feature.**

| <b>Range</b>                      | <b>No. of Responses</b> | <b>Percentage (%)</b> |
|-----------------------------------|-------------------------|-----------------------|
| Strongly Agree                    | 45                      | 80.4%                 |
| Agree                             | 11                      | 19.6%                 |
| Neutral                           | 0                       | 0%                    |
| Disagree                          | 0                       | 0%                    |
| Strongly Disagree                 | 0                       | 0%                    |
| <b>TOTAL</b>                      | 56                      | 100%                  |
| <b>MEAN:4.80 – Strongly Agree</b> |                         |                       |

*Table 14 Question No.8 Responses*

Table 14 presented the responses of 56 participants. Among them, 45 respondents, equivalent to 80.4%, strongly agreed with the statement and 11 respondents or 19.6% agreed. The responses had a mean score of 4.80, which fell within the range of strongly agreed. This result indicated that receiving alerts directly to the smartphone was a useful feature during an emergency.

**Question #9: The mobile app for monitoring the home seems user-friendly.**

| <b>Range</b>                      | <b>No. of Responses</b> | <b>Percentage (%)</b> |
|-----------------------------------|-------------------------|-----------------------|
| Strongly Agree                    | 35                      | 62.5%                 |
| Agree                             | 20                      | 35.7%                 |
| Neutral                           | 1                       | 1.8%                  |
| Disagree                          | 0                       | 0%                    |
| Strongly Disagree                 | 0                       | 0%                    |
| <b>TOTAL</b>                      | 56                      | 100%                  |
| <b>MEAN:4.61 – Strongly Agree</b> |                         |                       |

*Table 15 Question No.9 Responses*

Table 15 presented the responses of 56 participants. Among them, 35 respondents, equivalent to 62.5%, strongly agreed with the statement, while 20 respondents or 35.7% agreed, and 1 respondent or 1.8% was neutral. The responses had a mean score of 4.61, which fell within the range of strongly agreed. This result indicated that the mobile application ISHA was user-friendly.

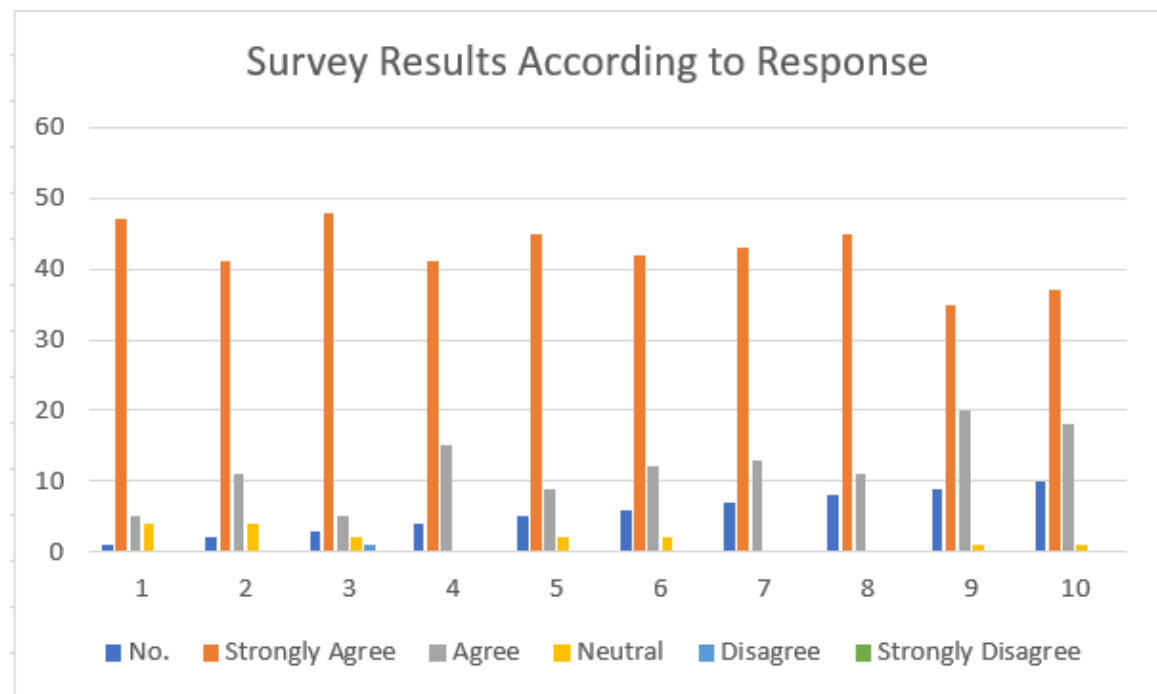
**Question #10: The way the system sends app alerts is reliable.**

| <b>Range</b>                      | <b>No. of Responses</b> | <b>Percentage (%)</b> |
|-----------------------------------|-------------------------|-----------------------|
| Strongly Agree                    | 37                      | 66.1%                 |
| Agree                             | 18                      | 32.1%                 |
| Neutral                           | 1                       | 1.8%                  |
| Disagree                          | 0                       | 0%                    |
| Strongly Disagree                 | 0                       | 0%                    |
| <b>TOTAL</b>                      | 56                      | 100%                  |
| <b>MEAN:4.64 – Strongly Agree</b> |                         |                       |

*Table 16 Question No.10 Responses*

Table 16 presented the responses of 56 participants. Among them, 37 respondents, equivalent to 66.1%, strongly agreed with the statement, while 18 respondents or 32.1% agreed, and 1 respondent or 1.8% was neutral. The responses had a mean score of 4.61, which fell within the range of strongly agreed. This result indicated that having an alert feature for the mobile application was reliable.

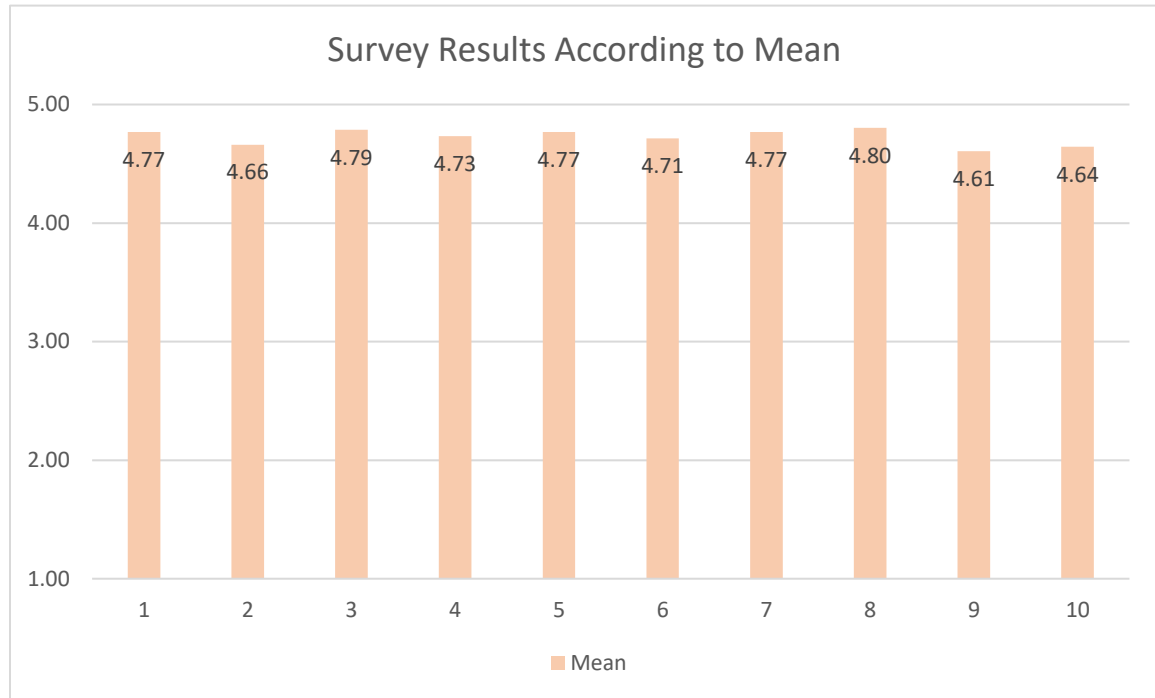
## Survey Results According to Response



*Figure 65 Survey Results According to Response*

The figure showed the graphical representation of the survey results according to the total number of respondents. The x-axis displayed the numbers 1 to 10, referring to ten post-survey questions. The y-axis indicated the total number of responses per question. The bars represented the total number of responses from five standard response categories such as Strongly Agree, Agree, Neutral, Disagree, and Strongly Disagree.

### Survey Results According to Mean



*Figure 66 Survey Results According to Mean*

The figure displayed the graphical representation of the post-survey results, where the x-axis represented the ten survey questions numbered 1 to 10, and the y-axis indicated the calculated range of the mean score for each corresponding question.